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From correlation to causation: data-mechanism dual-driven analysis of defects induced by melt pool dynamics, vapour recoil, and spatter in PBF-LB and mitigation

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ABSTRACT

Laser beam powder bed fusion (PBF-LB) is reshaping high-end manufacturing. However, the non-equilibrium metallurgical process involves transient phenomena governed by multiphysics coupling, and their correlation with build quality remains unclear. Consequently, defect control still relies on empirical trial-and-error. This paper analyzes the formation mechanisms of critical physical phenomena during PBF-LB. It assesses the principles, capabilities, and limitations of in situ monitoring techniques, including synchrotron X-ray, high-speed optical, schlieren, and acoustic emission. Mechanistic modelling approaches, spanning macroscale thermomechanical modelling, mesoscale melt pool dynamics, and microstructural evolution, together with their applications in defect prediction, are critically analyzed. Furthermore, data-mechanism dual-driven methods, integrating multimodal data and physics-informed machine learning, link melt flow, keyhole instability, vapour recoil, and spatter to porosity, lack of fusion, cracking, and segregation. Finally, recent advances in defect-mitigation strategies are analyzed, including process-window optimisation, spatiotemporal beam shaping, and multiphysical-field assistance, to support defect-free PBF-LB manufacturing.

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1. Introduction

Laser beam powder bed fusion (PBF-LB) selectively melts metal powder layer by layer using a high-energy laser beam, achieving geometric freedom and complexity unattainable by conventional subtractive or isothermal processes. This layer-wise, locally confined forming process is inherently an extreme non-equilibrium metallurgical procedure that couples thermal, mechanical, fluid, and phase fields across multiple temporal and spatial scales. In the micrometer-scale melt pool, temperature gradients reach 10^6 K/m, cooling rates range from 10^5 to 10^7 K/s, and melt flow velocities are on the order of m/s [1, 2]. Under these extreme conditions, multiple intertwined physical phenomena govern energy transfer, mass transport, and phase transformations. Marangoni convection, driven by surface tension gradients from temperature or compositional variations, is the dominant melt pool circulation mechanism, directly dictating convective mixing and heat transport. Vaporisation and recoil arise from the intense laser-metal interaction. The vapour jet produced by metal evaporation exerts recoil pressure on the melt pool surface, profoundly modifying melt pool geometry and stability. When the energy density exceeds a critical threshold, the vapour depression can evolve into a deep, narrow keyhole. The dynamic interfacial instability of this keyhole is governed by the coupling of vapour flow, liquid inertia, and surface tension fluctuations, directly influencing energy absorption efficiency and porosity formation [2, 3]. These phenomena operate throughout the entire additive manufacturing cycle and are central to understanding energy-matter interaction.

Understanding these phenomena and their link to defect formation has progressed from a black-box to a grey-box paradigm. Early investigations relied mainly on ex-situ characterisation of as-built parts, deducing

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defect origins through metallography and electron microscopy. Such ex-situ methods are inherently limited in revealing the dynamic processes and transient mechanisms behind defect formation. More recently, high-spatiotemporal-resolution in situ monitoring, especially synchrotron X-ray imaging, has allowed direct observation of keyhole oscillations, bubble nucleation, and spatter trajectories within the melt pool [4, 5]. Concurrently, multi-scale numerical modelling, spanning from macroscopic thermo-mechanical coupling to mesoscopic melt pool hydrodynamics and microstructural evolution, has supplied a powerful toolset for interpreting the physics behind experimental observations [6,7]. Building on these advances, data-driven methodologies now enable automatic identification of defect signatures from large monitoring datasets and the establishment of process-quality correlations [8–10].

Despite this progress, profound challenges persist. First, at the physical mechanism level, although individual phenomena such as Marangoni convection, recoil pressure, and keyhole dynamics have been studied in isolation, the coupling among multi-physics fields and multiphase interfaces under extreme transient conditions still lacks a unified theoretical framework. For example, quantitative description of melt pool turbulence transition, nonlinear keyhole interface evolution, and momentum and heat exchange between the vapour plume and powder particles remains highly dependent on case-specific numerical simulations, and universal physical models have yet to be established. Second, at the process monitoring level, despite the abundance of sensor data, fusing multi-source, heterogeneous, high-dimensional monitoring information and extracting features with strong causal links to defect formation remains a bottleneck for intelligent process surveillance. Finally, at the process control level, most current strategies are offline-optimised open-loop approaches, where a safe process window is predetermined through extensive trial-and-error experiments. Such strategies cannot accommodate dynamic instabilities induced by stochastic disturbances during the build, including powder layer thickness fluctuations, localised heat accumulation, or spatter re-deposition [11].

Recent studies have examined image-based monitoring and machine-learning-enabled control [12], layer-wise defect detection and localisation with convolutional neural networks [13], defect forecasting with time-series foundation models [14], and physics-guided multitask learning for inferring subsurface defects from top-view monitoring data [15]. These studies mainly address individual tasks such as detection, localisation, forecasting, or image analysis. In contrast, the present paper follows the physical sequence from energy input and material response to melt pool, vapour, keyhole, spatter, and denudation dynamics, and then to defect formation, observability, modelling, and intervention. It also distinguishes correlation-based prediction from mechanism-constrained interpretation and analyzes the validation requirements and engineering limitations associated with transferable process control.

To overcome these challenges, this paper systematically analyzes the formation mechanisms, characterisation methods, and control strategies of critical physical phenomena in PBF-LB, with particular emphasis on the intrinsic correlations among melt pool dynamics, vapour plume behaviour, spatter ejection, denudation, and the resulting typical defects. The state of the art in in situ monitoring techniques, multi-scale modelling approaches, and data-driven models is critically assessed. The paper further explores future directions toward intelligent and autonomous quality control, aiming to provide theoretical reference and methodological support for process optimisation and defect mitigation in additive manufacturing.

2. In situ monitoring methods

Understanding transient phenomena in PBF-LB requires measurements with sufficient temporal and spatial resolution. Some events occur within the metal, while others occur at or above the melt pool surface, and their characteristic times range from nanoseconds to microseconds. Monitoring methods must therefore be selected according to both their resolution and their ability to penetrate the material or access the relevant surface. This section analyzes methods that image the melt pool interior or capture surface and gas-phase dynamics. Their operating principles, measurement capabilities, advantages, and limitations are discussed.

2.1. Synchrotron X-ray imaging

Synchrotron X-ray imaging has been widely used to investigate PBF-LB. The high-brilliance, high-energy radiation can penetrate metal samples several millimetres thick. When combined with high-frame-rate, high-resolution detectors, this technique enables in situ observation with micrometer-to-submicrometer spatial resolution and microsecond-to-nanosecond temporal resolution. Synchrotron X-ray setups typically employ glassy carbon or beryllium windows to ensure X-ray transmittance and are equipped with fibre lasers, galvanometer scanners, and controlled atmosphere systems consistent with industrial configurations. These systems permit direct observation of melt pool

dynamics, defect formation, and powder behaviour under industrially relevant process parameters [4, 16–20].

Beamline and detector configurations provide different combinations of spatial resolution, temporal resolution, and field of view, as shown in Figure 1(a)-(c). For example, the ID19 beamline at the European Synchrotron Radiation Facility (ESRF) uses high-energy polychromatic X-rays (mean energy 30 keV) to achieve 4.3 µm pixel resolution at a 40 kHz frame rate, making it suitable for prolonged tracking of melt pool evolution and spatter behaviour during multilayer printing [17, 19]. The system built at beamline 32-ID-B of the Advanced Photon Source (APS) employs an undulator-generated beam (~24 keV, 47% bandwidth) coupled with a Photron high-speed camera, achieving up–1 MHz frame rate, 500 ns exposure time, and 1 µm pixel resolution. This setup can clearly capture keyhole fluctuations, powder spatter, and pore formation processes [4, 18, 22, 23]. More recently, the application of X-ray free-electron lasers has further pushed the temporal resolution to the nanosecond regime. For instance, the transmission X-ray microscope at the Linac Coherent Light Source (LCLS) in the United States, using compound refractive lenses and a two-pulse mode, has achieved an effective pixel size of 206 nm and a frame interval of 2.1 ns, allowing measurements of ultrafast phase transformations and material response under extreme conditions [2].

This penetrating imaging capability has provided direct evidence for mechanisms that were previously inferred. In situ observations have resolved the formation, periodic oscillation, and collapse of keyholes, the nucleation, growth, migration, and eventual entrapment of bubbles by the solidification front, the trajectories of unmelted powder particles entrained into the melt, and the transient melt flow velocity field reconstructed via tracer particle methods [21]. For example, ultrahigh-speed synchrotron X-ray imaging has captured in real time the full transition from conduction mode to keyhole mode, clearly delineating the energy density threshold for keyhole formation at approximately 0.4 MW/cm^2 and further suggesting that the keyhole front wall angle can serve as a criterion for assessing keyhole stability.

Beyond high-speed two-dimensional imaging, X-rays can also enable three-dimensional non-destructive characterisation of the PBF-LB process. The miniature PBF-LB setup developed at the ESRF ID19 beamline, incorporating a rotation stage and an sCMOS detector, allows tomographic scanning after each layer's powder deposition and melting, achieving a voxel size of 3.64 µm and thereby tracking the evolution of porosity defects across multiple printed layers [21, 24].

The method nevertheless has several constraints. First, there is a fundamental trade-off between high temporal resolution and field of view. Microsecond or nanosecond exposures often require a submillimeter field of view, preventing simultaneous observation of large-scale melt pool evolution and localised transient events. Second, penetration depth and spatial resolution are mutually constrained. Dense metals generally require higher-energy X-rays, which reduce image contrast and hinder the resolution of submicrometer bubbles or precipitates. Third, sample thickness and the number of build layers are restricted by X-ray attenuation. Thick samples and previously deposited layers

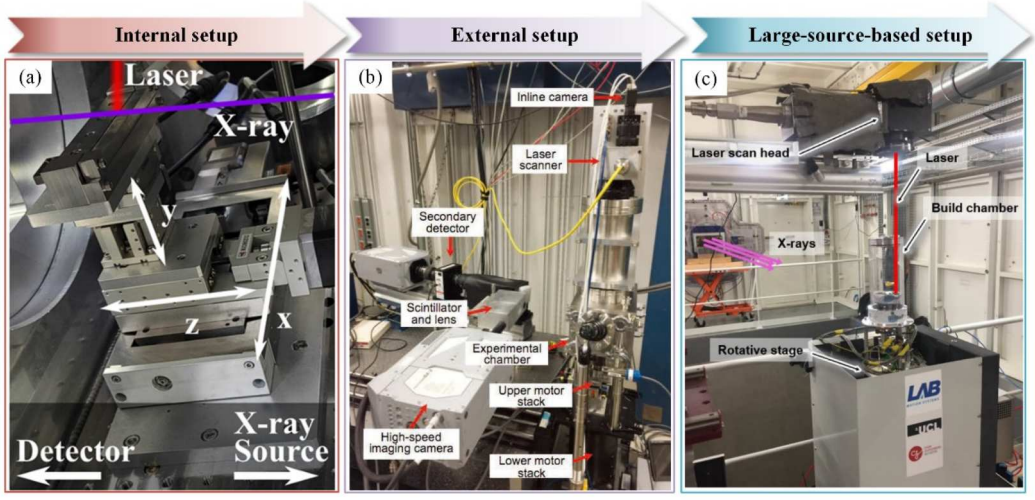


Figure 1. (a) X-ray system integrated inside a printer [2]. (b) Printing system modified with imaging setup [18]. (c) Printing apparatus constructed around a large-scale synchrotron source [21].

attenuate the signal, so experiments often use a single layer or only a few layers and cannot fully reproduce heat accumulation and defect evolution in industrial multilayer builds. These constraints make synchrotron X-ray imaging a primary tool for mechanism validation rather than a routine process monitoring technique.

2.2. High-speed optical imaging

Unlike X-ray imaging, high-speed optical imaging records phenomena at and above the melt pool surface. It is a conventional tool for studying melt pool morphology, temperature fields, spatter behaviour, and vapour plume dynamics. Depending on the observation target, this technique includes visible-light imaging and infrared thermal imaging, each with distinct configurations such as coaxial or off-axis stereo vision and with different spatiotemporal resolution characteristics.

In the visible-light domain, a coaxial high-speed CMOS imaging system couples the camera to the laser optical path, recording two-dimensional melt pool morphology at tens of thousands of frames per second (Figure 2(a)). By combining Sobel edge detection with physical models, melt pool width and length can be extracted online and correlated with melt pool depth [25]. Off-axis stereo vision extends these measurements to three dimensions. Using two or more synchronised cameras at different angles, together with epipolar geometry and triangulation, spatter trajectories can be reconstructed and used to calculate velocity, acceleration, and ejection angle [26]. A dual-camera stereo vision system (1000 fps, 1280 × 720 pixels) that synchronously images the PBF-LB process can detect more than

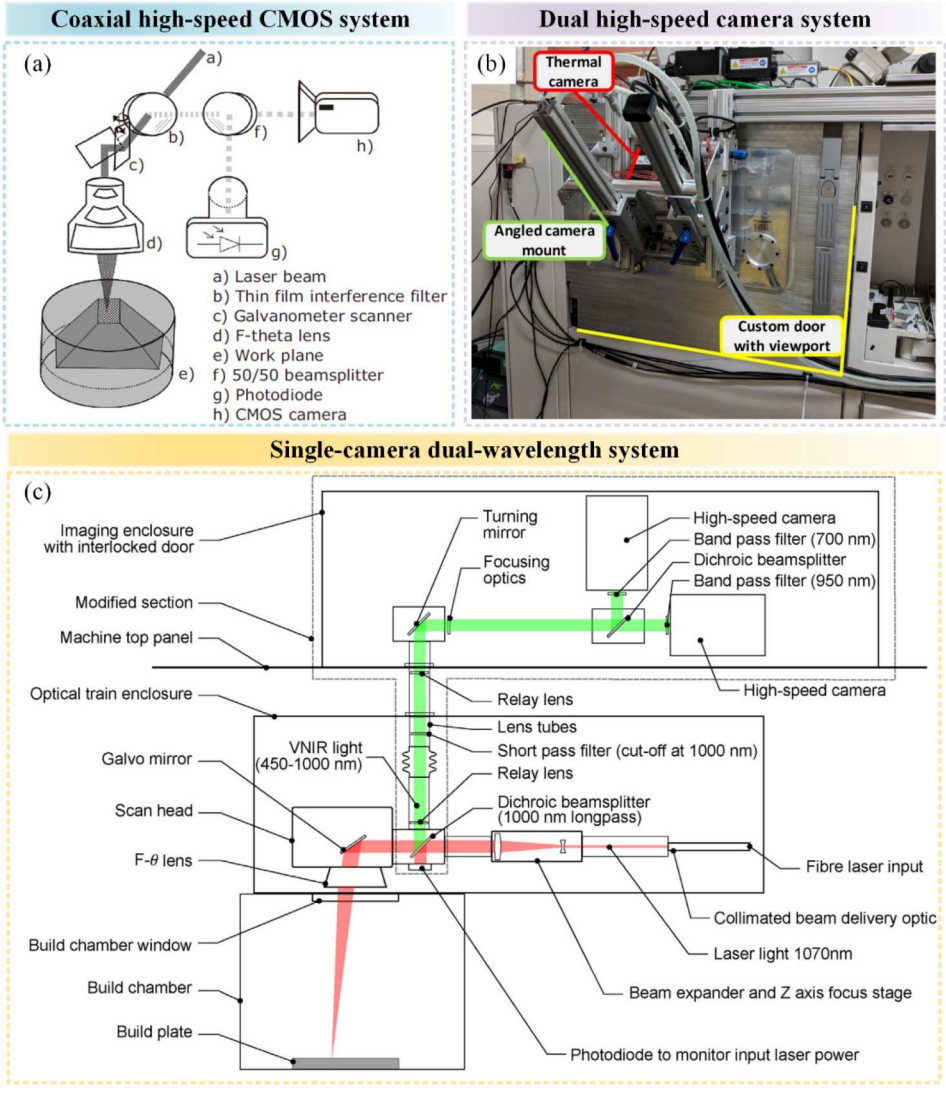


Figure 2. (a) Coaxial high-speed CMOS imaging system [25]. (b) Dual high-speed camera imaging system [27]. (c) Single-camera dual-wavelength system [28].

10000 spatter events and reveal the statistical distribution of spatter velocity and size [26].

Infrared imaging estimates the evolving surface temperature field from thermal radiation emitted by the melt pool. Early single-wavelength infrared imaging was limited by the uncertainty of material emissivity. Two-colour ratio pyrometry reduces this sensitivity to emissivity. By simultaneously collecting radiation images in two distinct wavelength bands (e.g. 700 nm and 950 nm) and using the relationship between the radiation intensity ratio and temperature, the influence of emissivity can be reduced, enabling high-spatiotemporal-resolution measurement of the melt pool temperature field. A dual high-speed camera system (100 kHz, 20 µm/pixel, see Figure 2(b)) employs this principle to capture dynamic melt pool temperature distributions [27]. A single-camera dual-wavelength system, realised through beam-splitting optics and chromatic aberration correction, can achieve the same objective (see Figure 2(c)) [28]. The temperature data can be used to calculate cooling rates, validate heat conduction models, and, through the thermal retention effect, identify lack-of-fusion defects. In infrared images, the obstructed heat dissipation path in an un-fused region manifests as a hotspot with abnormally elevated temperature.

High-speed optical systems are readily integrated and can provide rapid feedback, but their performance is limited by trade-offs among acquisition rate, field of view, and spatial resolution. Increasing the frame rate generally reduces the available field of view or spatial resolution. Achieving a temporal resolution of hundreds of thousands of frames per second typically requires a substantial reduction in the pixel array or a sacrifice of spatial resolution, making it difficult to simultaneously capture macroscopic melt pool morphology and micron-scale spatter details. Second, the interplay between exposure time and image brightness means that extremely short exposures must rely on intense melt pool radiation, which can easily cause image saturation, while reducing the exposure may lead to information loss in weak-signal regions such as the melt pool edge. The optical diffraction limit imposes an additional bound on spatial resolution. Even at high magnification, submicrometer resolution is difficult to achieve, and optical imaging cannot directly resolve microstructure evolution within the melt pool. These factors restrict the application scope of optical surface observation.

2.3. Schlieren imaging

The interaction between the metal vapour plume above the melt pool and the shielding gas flow is critical for understanding recoil pressure, powder denudation, and spatter driving forces. Because these gas flows are

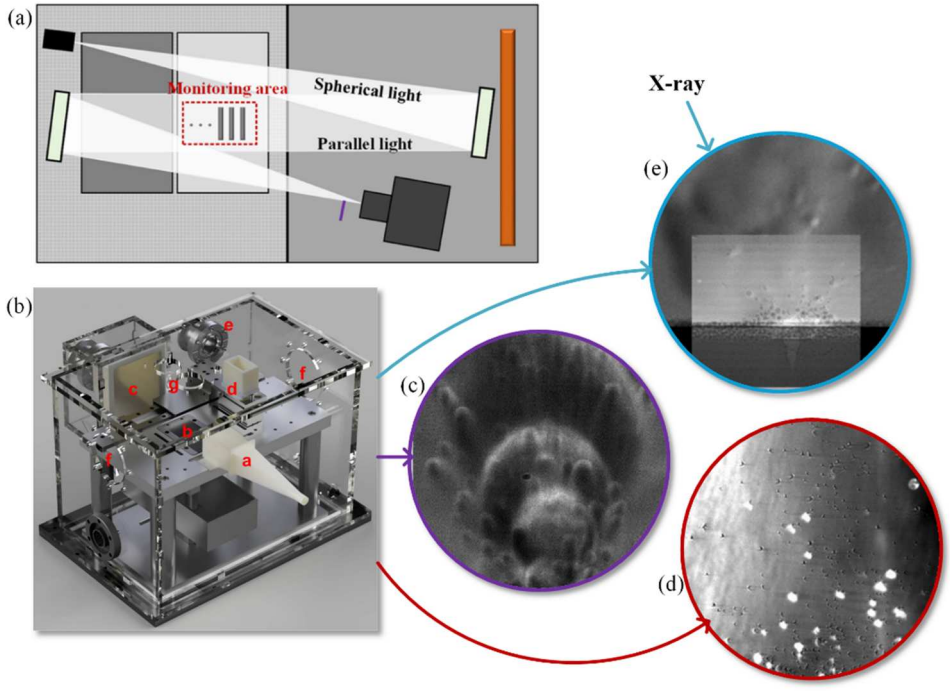


Figure 3. (a) Principle of the Z-type schlieren system [31]. (b) Schlieren monitoring system set up inside the build chamber [33]. (c) Images of vapour-induced spatter motion [30]. (d) Identification of cold and hot spatter [33]. (e) Simultaneous monitoring of melt pool, vapour plume, and spatter combined with an X-ray system [34].

transparent, conventional intensity imaging provides little contrast. Schlieren imaging instead detects density gradients and provides a nonintrusive view of gas-phase behaviour coupled to the liquid and solid phases. The method measures the deflection of light passing through refractive-index gradients. By placing a spatial filter such as a knife edge between the light source and the camera, the deflection is converted into image intensity variations, thereby visualising the density field that reflects temperature and concentration distributions within the transparent medium [29].

Various schlieren setups have been developed to meet different monitoring requirements. As shown in Figure 3(a, c), the Z-type schlieren system employs an off-axis reflective optical path and offers a relatively large field of view (several centimetres), making it suitable for observing the macroscopic morphology of the vapour plume and its interaction with the shielding gas flow [30, 31]. The lens-type schlieren system is more compact and easier to integrate into PBF-LB equipment [32]. The high-magnification schlieren system (Figure 3(b)) uses long-focal-length lenses and microscope objectives to achieve a spatial resolution of 15.6 µm/pixel, sufficient to resolve the morphology and trajectories of individual cold and hot spatter particles (Figure 3(d)) [33]. Schlieren imaging can also be combined with high-energy synchrotron X-ray imaging for simultaneous monitoring. Through a folding mirror, the optical beam is aligned at an angle of approximately 2° with the X-ray beam, enabling near-coaxial observation of the spatiotemporal correlations among keyhole depression, vapour plume, and spatter (Figure 3(e)) [34].

These schlieren observations have revealed key multiphase coupling mechanisms. The vapour jet angle has been found to increase significantly with scan speed. At low speeds the jet is nearly vertical, whereas at high speeds it can deflect to an obtuse angle or even be directed backward [31]. This change directly reflects the variation in melt pool front wall inclination and serves as an external indicator of keyhole mode transition. When schlieren imaging is synchronously combined with X-ray imaging, a causal link between keyhole instability and plume chaos is directly established. The periodic collapse of the keyhole rear wall synchronously induces violent oscillations in the vapour plume, and vice versa [34].

Schlieren images also support quantitative analysis. Optical flow methods have been used to calculate gas flow velocity and turbulence intensity [35]. A U-Net convolutional neural network has been trained to segment the plume and melt pool regions, enabling quantification of plume intensity, arm length, and deflection angle [31]. Background-oriented schlieren processing enhances sensitivity to weak refractive index gradients, allowing the detection of low-temperature vapour plumes [33]. These quantitative methods support correlations between gas-phase dynamics and defect formation.

Nevertheless, the spatiotemporal resolution of schlieren imaging is constrained by several optical factors. Temporal resolution is limited by the camera frame rate and luminous flux, and capturing microsecond-scale plume oscillations often requires reducing image resolution or the field of view. Spatial resolution is governed by the numerical aperture and alignment accuracy of the schlieren optical path. The narrow knife-edge cutoff needed for high sensitivity significantly reduces image illuminance, further limiting exposure time and signal-to-noise ratio. Furthermore, schlieren images represent a line-of-sight integration, essentially a two-dimensional projection of the density gradient along the optical path, which precludes direct differentiation of flow structures in the depth direction. Three-dimensional reconstruction would require multi-view imaging, multiplying the demands on optical access and synchronised triggering.

2.4. Acoustic monitoring

When the laser interacts with the material, melt pool oscillations, vapour jetting, keyhole fluctuations, spatter impacts, solidification shrinkage, and microcrack propagation radiate elastic stress waves that constitute acoustic emission (AE) signals [36–38]. As shown in Figure 4(a), these signals can be captured in a contact or non-contact manner by highly sensitive acoustic sensors placed inside the build chamber, and the measured signals can indicate changes in process state. Based on the sensing principle, the acoustic monitoring techniques currently applied to PBF-LB generally use either piezoelectric AE sensors or optical microphones.

Piezoelectric AE monitoring relies on the piezoelectric effect. Sensors are typically installed in contact with the build substrate or component surface, directly receiving elastic stress waves propagating through the solid medium. They have a broad frequency response (typically 20 kHz to 1 MHz) and are particularly sensitive to high-frequency transient signals generated by crack propagation and keyhole instability [36–38]. Because the sensors are closely coupled to the structure, signal attenuation is low and the signal-to-noise ratio is high, facilitating long-term continuous monitoring. However, this contact mounting approach is susceptible to thermal expansion in a high-temperature printing

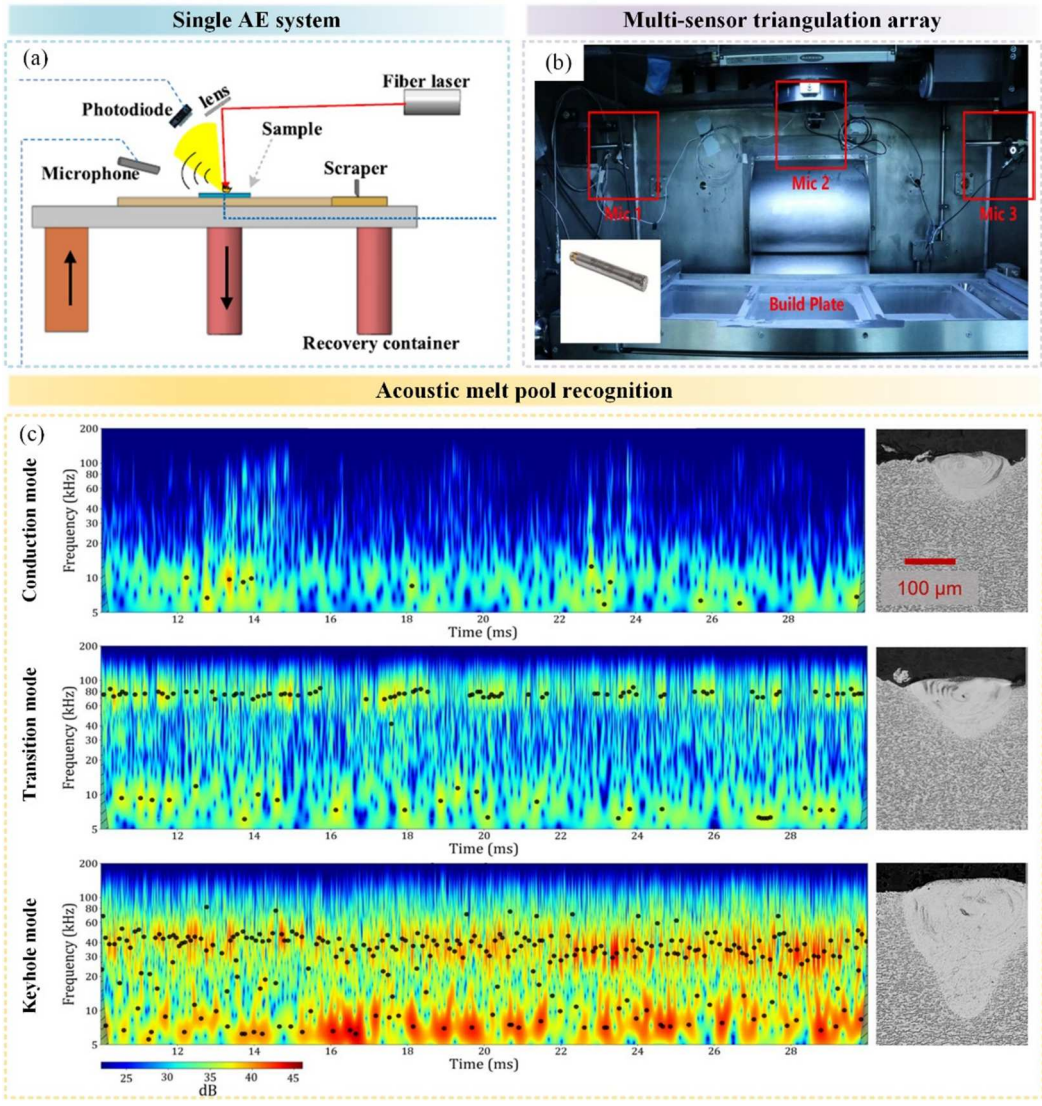


Figure 4. (a) A single AE monitoring system [38]. (b) A multi-sensor array triangulation system [39]. (c) Time-frequency characteristics of acoustic signals used to differentiate melting modes such as conduction mode, keyhole mode, and lack of fusion [36].

environment, and sensor placement is constrained by the substrate geometry. In a chamber environment with coupled multi-source noise, the accuracy of the wave velocity model can also directly affect the precision of arrival time picking and source localisation [39].

Optical microphones, in contrast, operate on a non-contact measurement principle, utilising laser interferometry or detection of refractive index changes induced by acoustic waves. The feasibility of applying such sensors to PBF-LB process monitoring has been validated. The sensing probe can be placed outside the build chamber and acquire signals remotely through a window [39]. A typical membrane-free optical microphone reconstructs acoustic signals by detecting gas refractive index variations caused by sound waves. It has a broadband response from infrasound to more than 2 MHz and is insensitive to electromagnetic interference, which is advantageous at high temperature, under vacuum, or in strong electromagnetic fields [40]. In PBF-LB, optical microphones can capture airborne acoustic signals from melt pool oscillations, vapour jetting, and keyhole fluctuations [36, 41]. Their non-contact nature also facilitates integration with robots or automated production lines [40]. Nevertheless, this technique is relatively sensitive to airborne dust, gas flow disturbances, and window contamination, and any interference along the signal propagation path can compromise measurement stability. Moreover, compared with piezoelectric sensors, the long-term operational validation of optical microphones remains insufficient [41].

As shown in Figure 4(c), time-frequency analysis of acoustic signals can distinguish melting states such as conduction mode, keyhole mode, and lack of fusion

[36, 42, 43]. Regardless of sensor type, acoustic monitoring spans multiple scales. In the time domain, sampling frequencies from 10 kHz to 1 MHz capture microsecond melt pool transients, and analysis window lengths from 12.5 ms [42, 44] to 0.5 s [45] balance temporal resolution and frequency domain resolution. Spatially, a single sensor cannot localise defects, but a multi-sensor array enables triangulation with millimeter-scale localisation error (Figure 4(b)) [39]. Acoustic signatures have been statistically linked to spatter behaviour [38, 46], porosity [43, 47], surface roughness [48], grain size [48], and support failure [48]. For optical microphones, convolutional neural networks have been applied to automatically extract and classify defect features such as pores and spatter [36, 41].

Combining sensors can improve the reliability of process monitoring. Acoustic signals are often combined with photodiode signals to aid defect classification [38, 47], with infrared thermal imaging for spatial localisation of AE sources [48], and with melt pool geometry images to complement surface and internal information [45]. Deep learning methods are widely employed for feature extraction and pattern recognition of acoustic signals, including convolutional neural networks [43, 46, 47], recurrent neural networks [46], extreme gradient boosting [42], and self-supervised learning [44], to identify frequency-band features associated with defects and to improve generalisation and interpretability [42, 47].

However, the spatiotemporal resolving capability of AE monitoring has intrinsic limitations. Although high sampling rates provide high temporal resolution, high-frequency acoustic waves attenuate significantly during propagation, constraining the effective detection range and sensor placement flexibility. Spatial localisation accuracy is restricted by the array geometry and the wave velocity model. In scenarios with complex component geometries and coupled multi-source noise, high-precision real-time source reconstruction remains difficult. More importantly, the time-frequency resolution of acoustic feature extraction is governed by the uncertainty principle, so high temporal resolution and high frequency resolution cannot be simultaneously attained to the same degree. This imposes a trade-off between precisely identifying microsecond transient events (e.g. instantaneous keyhole instability) and fully capturing broadband characteristics, and a single sensor cannot directly decouple the complex acoustic sources from superimposed multiple physical processes. Piezoelectric contact sensors may introduce additional constraints in high-temperature, high-speed scanning scenarios. Optical microphones avoid contact interference, but their sensitivity to environmental conditions and their long-term operational reliability still require further verification [40, 41]. Although their noncontact nature is useful in harsh environments, the issue of long-term stability must be addressed for engineering applications [40].

The above four monitoring methods complement each other. Table 1 compares their representative measurement ranges, and Table 2 relates those capabilities to specific applications. Synchrotron X-ray imaging remains the reference method for resolving internal keyhole and pore dynamics, although it is generally unsuitable for routine production monitoring. High-

Table 1. Quantitative comparison of representative in situ monitoring modalities for PBF-LB.

Modality	Temporal resolution / acquisition rate	Spatial resolution / field of view	Measured or inferred quantities	Principal limitations	Refs
Synchrotron X-ray imaging	40 kHz frame rate at ESRF; up–1 MHz with a 500 ns exposure at APS; 2.1 ns inter-pulse interval at LCLS	4.3 µm/pixel at ESRF; 1 µm/pixel at APS; 206 nm effective pixel size at LCLS; 3.64 µm tomography voxel size	Internal keyhole geometry, bubble nucleation/migration, pore entrapment, tracer-based melt flow, powder motion	Sub-millimeter field of view at the highest rates; sample thickness and layer count are restricted; facility access	[4, 17–19, 22, 23]
High-speed visible-light and thermal imaging	Typical frame rate: 10^3–10^5 frames/s; up–100 kHz demonstrated for two-colour thermography	Stereo-vision example: 1280 × 720 pixels at 1000 frames/s; two-colour example: 20 µm/pixel	Surface melt pool dimensions, thermal radiance or calibrated temperature, cooling rate, spatter size, velocity, acceleration, and ejection angle	Trade-off among frame rate, field of view, and spatial resolution; detector saturation; emissivity and calibration uncertainty; occlusion; no direct subsurface view	[25–28]
Schlieren imaging	Camera-limited; microsecond plume oscillations require reduced field of view or resolution	Field of view of several centimetres for Z-type systems; 15.6 µm/pixel for high-magnification systems; ∼2° alignment demonstrated with X-ray	Vapour-plume boundary and deflection angle, density-gradient structures, optical-flow-derived velocity, turbulence proxies, and cold/hot spatter signatures	Line-of-sight integration prevents direct depth resolution; high magnification reduces illumination and signal-to-noise ratio	[30–35]
Acoustic monitoring	Typical sampling rate: 10 kHz–1 MHz; piezoelectric sensor response: 20 kHz–1 MHz; optical microphones: >2 MHz	Single sensor: no localisation; arrays: millimeter-scale localisation error; analysis windows ∼12.5 ms–0.5 s	Broadband acoustic signatures associated with keyhole fluctuation, spatter impact, crack initiation, porosity-related states, and support failure	Source superposition, propagation attenuation, geometry-dependent wave velocity, and the time-frequency resolution trade-off	[36–48]

Table 2. Application-oriented benchmarking and selection of in situ monitoring methods.

Monitoring objective	Preferred primary modality	Recommended complementary measurement	Key quantitative outputs	Rationale and principal caveat
Validate keyhole-to-pore causality	Synchrotron X-ray imaging	Coaxial optical or schlieren imaging	Keyhole depth/front-wall angle, collapse frequency, bubble trajectory, pore capture probability	X-ray resolves the subsurface event; optical/schlieren signals establish whether a production-compatible proxy exists.
Track spatter generation and transport	Stereoscopic high-speed imaging	Schlieren imaging or shielding-gas flow diagnostics	Particle size, 3D velocity, ejection angle, flight time, landing position, plume angle	Optical tracking measures particles, while schlieren or gas-flow data distinguish initial vapour forcing from downstream transport.
Measure melt pool temperature and morphology	Coaxial high-speed imaging and two-colour pyrometry	Ex situ metallography or X-ray computed tomography	Length, width, area, aspect ratio, peak temperature, cooling rate, uncertainty interval	Surface signals can be acquired online, but melt pool depth and absolute temperature require calibration and uncertainty reporting.
Detect cracking or abrupt instability	Piezoelectric acoustic-emission sensing or optical microphone	Thermal or visible-light imaging with build-coordinate registration	Event time, band energy, spectral centroid, source location, confidence score	Acoustics offers high temporal bandwidth; imaging and registration are required to localise the event and assign a physical cause.
Production closed-loop control	Coaxial optical or thermal imaging, or photodiode sensing	Acoustic sensing, if integration and processing latency are acceptable	Feature latency, inference latency, false-alarm rate, missed-detection rate, calibration drift	The deployable choice is governed by latency, robustness, optical-window contamination, data bandwidth, and actuator response rather than laboratory resolution alone.

speed optical and thermal imaging are better suited to coaxial, high-throughput observation of surface phenomena. Schlieren imaging is sensitive to gas-density gradients and vapour plume deflection, whereas acoustic monitoring offers broad temporal bandwidth but limited direct spatial information. Studies that seek to establish causality should combine synchronised methods because no single modality can observe the condensed phase, the gas phase, and internal defects at the same time.

3. Mechanistic models

Monitoring identifies features associated with a physical process, while simulation models are needed to interpret those features and predict their process outcomes. Simulations at the part, melt pool, and microstructure scales, together with surrogate and machine-learning models, are analyzed and discussed. These approaches provide different but complementary links among process conditions, transient phenomena, defects, microstructure, and performance.

Numerical simulation reveals physical fields (temperature, stress, flow) that are difficult to measure directly and validates mechanistic hypotheses. According to scale and objective, PBF-LB simulations are divided into the macroscopic part scale, the mesoscopic melt pool scale, and the microscopic microstructure scale (Figure 5). Macroscopic models simulate thermo-mechanical coupling and residual stress for process window screening and deformation control. Mesoscopic models resolve melt pool-powder interactions, keyhole dynamics, spatter, element distribution, and pores, providing mechanistic support for defect mitigation. Microscopic models address phase transformations and microstructural evolution to establish process-microstructure-performance relationships.

3.1. Macroscopic scale models

Macroscopic-scale models adopt a continuum assumption, treating the powder bed as a homogeneous material and solving temperature, stress, and deformation fields for the entire part. The objective is to predict residual stress, warping, and support forces, thereby suppressing cracking and deformation by adjusting scanning strategy and preheating temperature. Based on thermo-elasto-plasticity, a thermal analysis first solves the transient heat conduction equation. The resulting temperature history is then applied as a thermal load in the mechanical analysis to compute stress and deformation. The heat conduction equation, which incorporates latent heat of phase transformation, is [56]

$$\rho c \frac{\partial T}{\partial t} + \rho L \frac{\partial f_{pc}}{\partial t} = \nabla \cdot (k \nabla T) + Q_{vol} \tag{1}$$

where ρ represents the material density, c the specific heat capacity, k the thermal conductivity, T the temperature, t the time, L the latent heat of phase transformation, and f_{pc} the phase transformation fraction. The term Q_{vol} on the right-hand side denotes the volumetric laser heat source, which is often described by a Gaussian distribution model to represent the volumetric absorption process of laser energy within the powder layer.

To improve computational efficiency, several simplifications are adopted. The powder bed is treated as a

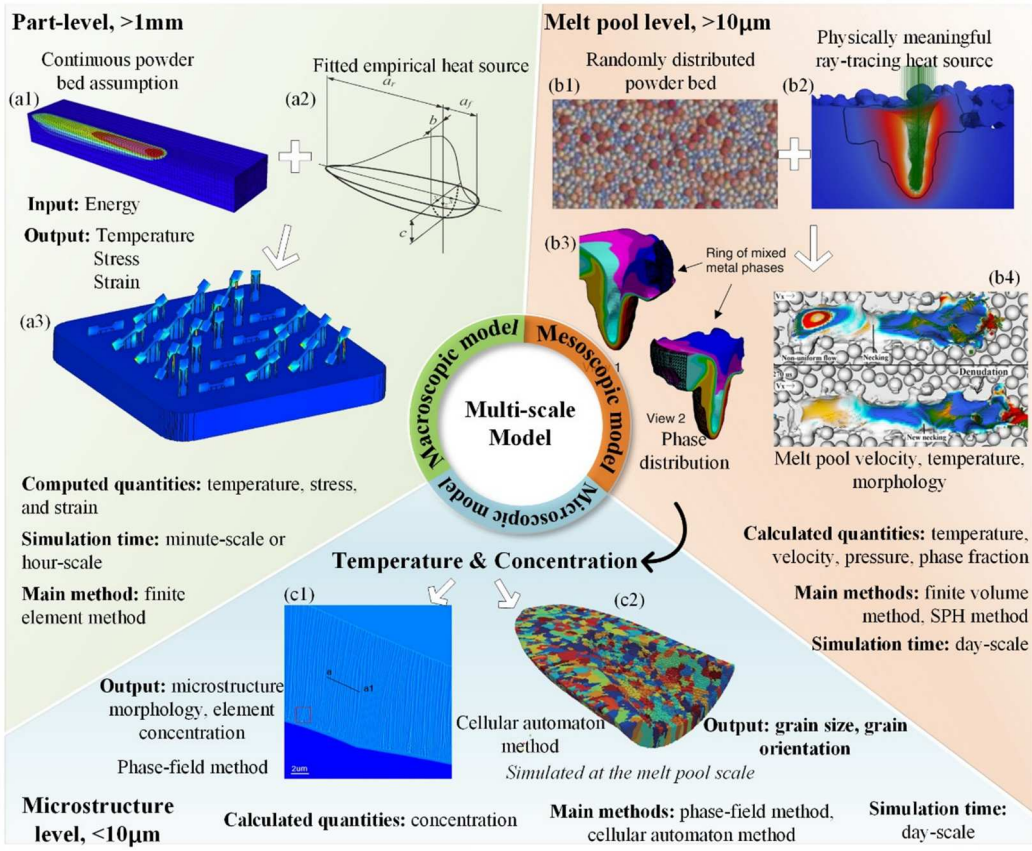


Figure 5. (a1) Continuum assumption of the powder bed in macroscopic models [49]. (a2) Empirical heat source fitted based on melt pool shape [50]. (a3) Simulation of part temperature, stress, and strain using a macroscopic model [51]. (b1) Randomly distributed powder bed in mesoscopic models [52]. (b2) Ray-tracing heat source designed based on laser-metal interaction [53]. (b3) Simulation of elemental distribution by adding phase properties in mesoscopic models [53]. (b4) Mesoscopic simulation results: temperature, velocity, pressure, and morphology [52]. (c1) Simulation of microstructure morphology and elemental segregation using the phase-field method, coupled with mesoscopic results [54]. (c2) Simulation of grain size and orientation at the melt pool scale using the cellular automaton method [55].

continuous homogeneous medium, and convective heat transfer from melt flow is neglected, considering only heat conduction (Figure 5(a1)). The heat source is calibrated by fitting a Gaussian volumetric source to the melt pool morphology (Figure 5(a2)). In the mechanical stage, thermal expansion and contraction derived from nodal temperature histories produce thermal stress because steep gradients constrain expansion. When this stress exceeds the temperature-dependent yield strength, plastic strain develops, determining residual stress and macroscopic distortion [56, 57]. Total strain comprises elastic $\{\varepsilon_e\}$, plastic strain $\{\varepsilon_p\}$, thermal strain $\{\varepsilon_{th}\}$, and phase transformation-induced strain $\{\varepsilon_{tr}\}$. The incremental constitutive relation is

$$\{d\sigma\} = [D_e](\{d\varepsilon\} - \{d\varepsilon_p\} - \{d\varepsilon_{th}\} - \{d\varepsilon_{tr}\}) \qquad (2)$$

where $\{d_\sigma\}$ is the stress increment and $[D_e]$ is the elastic stiffness matrix. The thermal strain increment is determined by the thermal expansion coefficient α, expressed as $\{d\varepsilon_{th}\} = \{\alpha\}\, d_T$. Solving these equations yields residual stress and deformation. The definition of the heat-affected zone directly affects mechanical boundary condition accuracy, and its size and temperature gradient serve as key crack susceptibility indicators [57].

The primary goal of thermo-mechanical simulation is process optimisation and defect mitigation. It identifies process parameters and scanning strategies that minimise residual stress (Figure 5(a3) [51]) and predicts distortion and cracking, allowing optimisation of part orientation and support structures before printing to prevent build failure [58]. More recent models also address microstructure evolution [56].

3.2. Mesoscopic scale models

Mesoscopic models simulate the multiphase interactions among powder particles, molten metal, and the ambient gas phase. The computational domain spans micrometers to millimetres, covering a few scan tracks or powder layers, and resolves velocity, pressure,

temperature, and interface morphology. These models capture transient melt pool evolution, keyhole dynamics, spatter formation, and pore entrapment, revealing the physical origins of defects such as gas pores, lack of fusion, and denudation zones and providing a mechanistic basis for refining process parameters. As shown in Figure 5(b1), unlike macroscopic finite element approaches that treat the powder as a continuum, mesoscopic simulations use the discrete element method to generate a powder bed with realistic size distribution and random packing. This powder bed geometry is then imported into a computational fluid dynamics model as the initial configuration, allowing the direct influence of local powder arrangement on melt pool morphology to be analyzed [59]. Mesoscale melt pool dynamics are governed by mass, momentum, and energy conservation equations. Source terms representing thermocapillary (Marangoni) convection, vapour recoil pressure, and mushy-zone resistance are incorporated into the momentum equation to accurately describe the melt flow [3, 60].

For the heat source, the ray tracing model has replaced the conventional volumetric assumption (Figure 5(b2)). By tracking multiple reflections and absorption, it captures the non-uniform laser heating of particle surfaces and the shadowing effect, providing the physical basis for modelling partially melted particles and the lateral shallow pores they induce [3, 61]. The ray tracing heat source is expressed as.

$$\dot{q}_{laser}(r) = \sum_{i=1}^{n_{ray}} \alpha_i I_0 \cdot \delta(r - r_i) \tag{3}$$

where α_i denotes the absorptivity of the i-th ray at its point of incidence, I_0 the laser intensity, and δ the Dirac delta function. This model reproduces the multi-reflection absorption phenomenon inside the keyhole, thus yielding more accurate energy distributions and cavity profile evolution under deep-penetration melting conditions. The volume of fluid method tracks the transient evolution of the melt pool free surface. Combined with recoil pressure and Marangoni convection, it reproduces the depression zone beneath the laser, the high-velocity melt ejection, and the recirculation flow structure in the rear part of the melt pool [60, 61].

Mesoscale models reveal transient defect formation mechanisms that are difficult to observe experimentally. Regarding keyhole-induced pores, high-spatiotemporal-resolution simulations have shown that galvanometer scanner acceleration and deceleration at turning points sharply increase local energy density, creating a deep keyhole depression that collapses within microseconds and entrains the shielding gas. This mechanism directly supports the formulation of power modulation strategies [60]. For powder denudation, simulation results indicate that under atmospheric pressure, the ambient gas shear flow induced by the metal vapour jet dominates powder entrainment and displacement, whereas under reduced pressure, direct momentum transfer from the vapour pushes the particles outward, explaining the non-monotonic variation of denudation zone width with ambient pressure [62]. Multi-material simulations can incorporate phase-specific properties and transport equations to reproduce material or elemental distributions (Figure 5(b3)) [53]. Mesoscopic simulations have also been extended from single-track to multitrack and multilayer scenarios, revealing asymmetric melt pool expansion induced by heat accumulation and the probability of defect formation in track overlap zones under different scanning strategies [59, 63].

Quantitative outputs of mesoscopic simulations, such as the temperature gradient and solidification rate at the solid-liquid interface, directly bridge to microstructure prediction models. This enables cross-scale prediction from process parameters to columnar grain morphology and even primary dendrite arm spacing, validating the applicability of solidification theory under high cooling rates [59]. These advances make mesoscale simulation useful for understanding defect origins, optimising process windows, and linking macroscopic thermo-mechanical response with microstructural evolution.

3.3. Microscopic scale models

Microstructure evolution directly determines mechanical properties, making high-fidelity simulation of phase transformation and grain growth essential. Simulations at this scale focus on individual grains, dendrites, and dislocation structures, solving for grain orientation, phase fraction, solute concentration, and dislocation density. They primarily predict the columnar-to-equiaxed transition (CET), dendrite arm spacing, eutectic phase distribution, and final mechanical properties, establishing quantitative process-microstructure-property relationships and offering tools for targeted microstructure control. Various numerical methods have been developed, among which the phase-field and cellular automaton methods [64] are the most widely used, while machine learning acceleration provides new avenues to overcome computational bottlenecks.

As shown in Figure 5(c1), the phase-field (PF) method, grounded in thermodynamics and kinetics, describes grain boundary evolution, solid-liquid interface migration, and competitive grain growth. An order parameter (e.g. solid fraction) captures the diffuse interface,

eliminating explicit interface tracking. For anisotropic grain growth, the governing equation derived from Ginzburg-Landau theory is [65]

$$\frac{\partial \phi}{\partial t} = M_\phi \left[\nabla \cdot (\epsilon^2 \nabla \phi) - \frac{\partial f(\phi, c, T)}{\partial \phi} \right] \tag{4}$$

where M_ϕ is the interface mobility, ε is the gradient energy coefficient related to the interfacial energy and anisotropy, and f is the bulk free energy density function, which couples the solute field c and the temperature field T. This model reproduces dendrite arm coarsening, competitive growth, and the CET with high fidelity and is particularly effective in capturing the preferred growth direction dictated by interfacial energy anisotropy. The high computational cost limits conventional PF models to small scales. To overcome this, a surrogate model based on a three-dimensional UNet convolutional neural network was proposed [65]. It takes the initial grain structure and temperature field as input, predicts grain orientation evolution, and employs a spatiotemporally composable strategy over local regions to avoid the exponential demand of full-scale simulation.

The cellular automaton (CA) method achieves a favourable balance between physical fidelity and computational efficiency and has been widely applied to grain morphology prediction at the mesoscopic scale [66–68]. The CA model couples the computed temperature field with nucleation and growth kinetics to predict solidification structure. The growth rate depends on local undercooling ΔT, and dendrite tip kinetics is typically described by the Kurz-Giovanola-Trivedi (KGT) or Lipton-Glicksman-Kurz (LGK) model [68].

$$v(\Delta T) = a_2 \Delta T^2 + a_3 \Delta T^3 \tag{5}$$

where a_2 and a_3 are material-dependent coefficients. The KGT model correlates the dendrite tip growth rate with the undercooling and is capable of predicting the primary dendrite arm spacing (PDAS) at extremely high cooling rates (10^5-10^7 K/s), which is a key characteristic that distinguishes PBF-LB from conventional casting [64]. In the CA model, grain growth of a cell is achieved by capturing adjacent liquid cells, and the growth length is calculated by integrating over the growth time step Δt.

$$l = \int_t^{t+\Delta t} v(\Delta T) dt \tag{6}$$

When a liquid cell is captured, it inherits the crystallographic orientation of the neighbouring grain, realising epitaxial growth. For Al-Si alloys, the CA model can switch between dendritic and eutectic growth modes, enabling simulation of grain refinement by laser remelting, solute redistribution, and sub-micron eutectic silicon network formation [68]. A two-dimensional coupled cellular automaton-finite element model has been used to investigate the influence of heterogeneous nucleation density on grain texture, clarifying how nucleation density determines the number of layers traversed by columnar grains and the grain size distribution [67]. Coupling a ray-tracing finite volume method (mesoscopic) with cellular automaton and incorporating thermocapillary effects further allows analysis of how the surface tension temperature coefficient affects grain size and morphology in different melt pool modes (Figure 5(c2)) [69].

Recent advances have introduced dislocation density as a key state variable. Coupling crystal plasticity or discrete dislocation dynamics with phase-field/CA methods enables direct simulation of thermal-stress-induced dislocation multiplication, motion, and entanglement, allowing prediction of yield strength and work hardening behaviour [70]. Such coupled thermal-phase-field-mechanical simulations enable quantitative analysis of dislocation evolution and microcrack nucleation under the steep temperature gradients in PBF-LB.

3.4. Models for material, geometry and build settings

Model fidelity depends on numerical resolution and on whether the input data represent the material and build configuration of interest. Relevant material inputs include temperature- and phase-dependent density, heat capacity, thermal conductivity, latent heat, viscosity, surface tension and its dependence on temperature and composition, absorptivity, emissivity, saturation vapour pressure, evaporation enthalpy, and alloy-species transport. Powder size distribution, morphology, packing density, oxide state, and interparticle contact affect optical absorption, capillary infiltration, and denudation. Constant room-temperature properties may be adequate for qualitative comparisons, but their use across different materials or energy regimes can shift the predicted melt pool dimensions, recoil pressure, Marangoni flow direction, spatter momentum, and solidification rate.

Geometry changes the local and global boundary conditions. At the melt pool scale, free-surface curvature, powder packing, track overlap, scan turns, and previously consolidated material affect laser absorption and fluid flow. At the component scale, thin walls, corners, overhangs, supports, and build height alter heat-loss paths, stiffness, residual stress, and acoustic-wave propagation. A model calibrated against a straight single track should therefore not be applied to hatch

Table 3. Model inputs and validation requirements for assessing transferability across materials, geometries, and build conditions.

Transferability factor	Minimum model inputs	Affected predicted outcomes	Recommended validation
Material and alloy composition	Temperature- and phase-dependent thermophysical properties; surface tension and its temperature coefficient; absorptivity; vapour pressure; species-transport parameters	Melt pool size and flow direction; keyhole depth; recoil pressure; selective evaporation; segregation	Independent validation for each material or composition and energy regime; uncertainty or sensitivity analysis for poorly known high-temperature properties
Powder-bed state	Particle-size distribution, morphology, packing, layer thickness, oxide/contamination state, contact and adhesion parameters	Absorption, lack of fusion, entrainment, denudation width, spatter size, and spatter count	Validation against measured powder statistics and repeated powder beds or layers, rather than a single stochastic realisation
Geometry and scan path	Track overlap, turns, contours, overhangs, supports, local wall thickness, prior-layer thermal history	Heat accumulation, local melting-mode transitions, distortion, lack of fusion, and acoustic propagation	Feature-specific tests at corners, overlaps, thin walls, and overhangs; multitrack/multilayer comparison
Shielding-gas flow and chamber configuration	Gas species and pressure; inlet/outlet velocity profile; scan-to-flow angle; obstructions; plume-flow coupling	Plume deflection, spatter removal or deposition, denudation, oxidation, and cross-contamination	In-chamber velocity or plume measurement, or chamber-scale CFD; perturbation of flow rate, direction, or pressure

intersections, contour scans, overhangs, or multilayer components without further validation. A hierarchical approach can transfer part-scale thermal histories and geometry-dependent boundary conditions to local high-fidelity simulations, which should then be tested at representative geometric features.

Conditions within the build chamber are also important. Shielding-gas composition, pressure, inlet velocity, scan direction relative to the gas flow, chamber obstructions, filter loading, and plume-induced recirculation affect spatter transport and powder denudation [62, 71–74]. A uniform velocity imposed at the boundary of a small computational domain can reproduce nominal cross-flow, but it cannot capture machine-scale recirculation or local velocity deficits behind components. Models of spatter and denudation should therefore obtain their local boundary conditions from chamber-scale computational fluid dynamics or measured velocity fields. Validation should include an independent material or powder condition, a change in geometry or scan path, and a change in gas flow or pressure. Otherwise, the conclusions should be presented as specific to the conditions studied. Finally, Table 3 summarises the model input and verification requirements for transferability among materials, geometries, and build conditions.

Macroscopic, mesoscopic, and microscopic models provide complementary information. Macroscopic models are computationally efficient but cannot resolve transient heterogeneity within the melt pool. Mesoscopic and microscopic models describe defect formation and microstructure evolution in greater detail, but their computational cost limits their use for broad parameter searches and routine simulations of multiple tracks, layers, or entire components. Predictive accuracy also depends on reliable material properties and heat-source calibration. Powder, melt, and vapour interactions, together with temperature-dependent properties and stochastic microstructures, make full-physics prediction impractical for real-time use. Data-driven surrogate models address this limitation by predicting thermal histories, defect distributions, microstructures, or properties at much lower computational cost.

4. Data models

4.1. Surrogate models

Surrogate models approximate the input-output behaviour of a costly experiment or high-fidelity simulation at a fraction of the computational cost. By learning from a designed set of representative samples, they enable rapid prediction, process-window exploration, optimisation, and uncertainty analysis. The following subsections distinguish statistical surrogates from reduced-order models and physics-informed approaches.

4.1.1. Surrogate models based on statistical learning

Statistical learning surrogate models establish explicit functional relationships between process parameters and key responses through a limited number of experiments or high-fidelity simulation samples, enabling rapid process window exploration and optimisation at very low computational cost. Such methods typically rely on design of experiments strategies, collecting sample data within a predefined parameter space and then employing regression analysis or Gaussian processes to construct the surrogate model. Based on the underlying modelling principles, they can be categorised into three typical approaches, namely response surface methodology, Kriging models, and dimensional analysis [75, 76].

Response surface methodology is the most fundamental and widely used statistical surrogate approach. It adopts full factorial or fractional factorial experimental designs to collect samples within the parameter space constituted by laser power, scan speed, and powder layer thickness. Explicit polynomial functions between responses such as melt pool width, depth, and morphological continuity and the parameters are then fitted through multivariate nonlinear regression. For the powder bed fusion process, researchers analyzed the influence of the three aforementioned parameters on melt pool morphology and dimensions via full factorial experiments, constructed a nonlinear regression model, and identified the optimal process parameter window [77]. The advantage of response surface methodology lies in its straightforward principle and transparent computation, while its prediction accuracy is limited by the polynomial order and it cannot quantify prediction uncertainty.

To overcome the limitations of response surface methodology, the Kriging model (also known as Gaussian process regression) introduces a spatial correlation structure that provides both a prediction mean and a prediction variance. This approach treats the response as a Gaussian random process, characterises the spatial correlation among sample points through a kernel function, and estimates model hyperparameters via Bayesian inference. In a porosity prediction study for stainless steel, researchers applied the spatial Gaussian process regression method with laser power and scan speed as inputs, achieving accurate porosity prediction using only a small number of initial experimental samples [78]. Through adaptive sampling strategies such as the maximum expected improvement criterion, new sample points can be targeted and added during iteration, continuously reducing prediction error at minimal experimental cost. The Kriging model reveals the physical influence patterns of the interaction between power and speed on porosity, and the prediction confidence intervals it provides offer decision support for risk-controllable process window optimisation.

Dimensional analysis provides a route to surrogate modelling through physics-based dimensionality reduction. The method groups multiple original process parameters (such as laser power, spot diameter, scan speed, and thermal diffusivity) into one or two dimensionless numbers through dimensional analysis, which maps the high-dimensional parameter space onto a low-dimensional feature space before an analytical regression model is fitted. Studies have shown that in the powder bed fusion process, the macroscopic melt pool behaviour is primarily controlled by a composite parameter, the ratio of laser power to spot diameter [79]. Based on this ratio, researchers established nonlinear mapping relationships with responses such as process efficiency, porosity, melt pool depth-to-width ratio, and growth ratio, and defined the quantitative range of the optimal process window [79]. The approach reduces the process space by identifying physically relevant parameter groups, yielding a parsimonious response surface model with extrapolation capability.

Statistical learning-based surrogate models use experimental or simulation data to approximate process responses with low-dimensional mappings. Response surface methodology is simple and intuitive, making it suitable for preliminary process window exploration. Kriging provides predictions and uncertainty estimates for small datasets, which is appropriate for optimisation tasks demanding higher accuracy. Dimensionless groups reduce the number of inputs while retaining a physical interpretation that can support extrapolation. The choice among these approaches depends on sample size, accuracy requirements, and the required level of interpretability.

4.1.2. Reduced-order models

Reduced-order models (ROMs) aim to predict full-field physical quantities such as temperature and stress at very low computational cost. They compress high-dimensional physical field data into a low-dimensional latent space and construct an end-to-end mapping from process parameters to the full field, achieving substantial acceleration compared with direct numerical simulation. This low cost makes ROMs particularly suitable for online monitoring and real-time feedback in PBF-LB. Based on the compression strategy, ROMs are divided into linear and nonlinear types, with the latter showing distinct advantages for strongly nonlinear problems.

Linear ROMs rely mainly on proper orthogonal decomposition (POD). Singular value decomposition is applied to snapshot matrices of temperature or stress fields from high-fidelity simulations, and the dominant energy modes are extracted. A mapping between process parameters and the POD coefficients is then built. For PBF-LB, a non-intrusive model combining POD with radial basis functions was proposed for online temperature field prediction. It incorporates laser power, scan speed, layer thickness, and material type, and is trained using a nonlinear material model with temperature-dependent properties (Figure 6(a)). The online prediction time was below 40 ms, meeting real-time requirements [80].

Linear models struggle with strong nonlinearity or discontinuities, prompting the development of

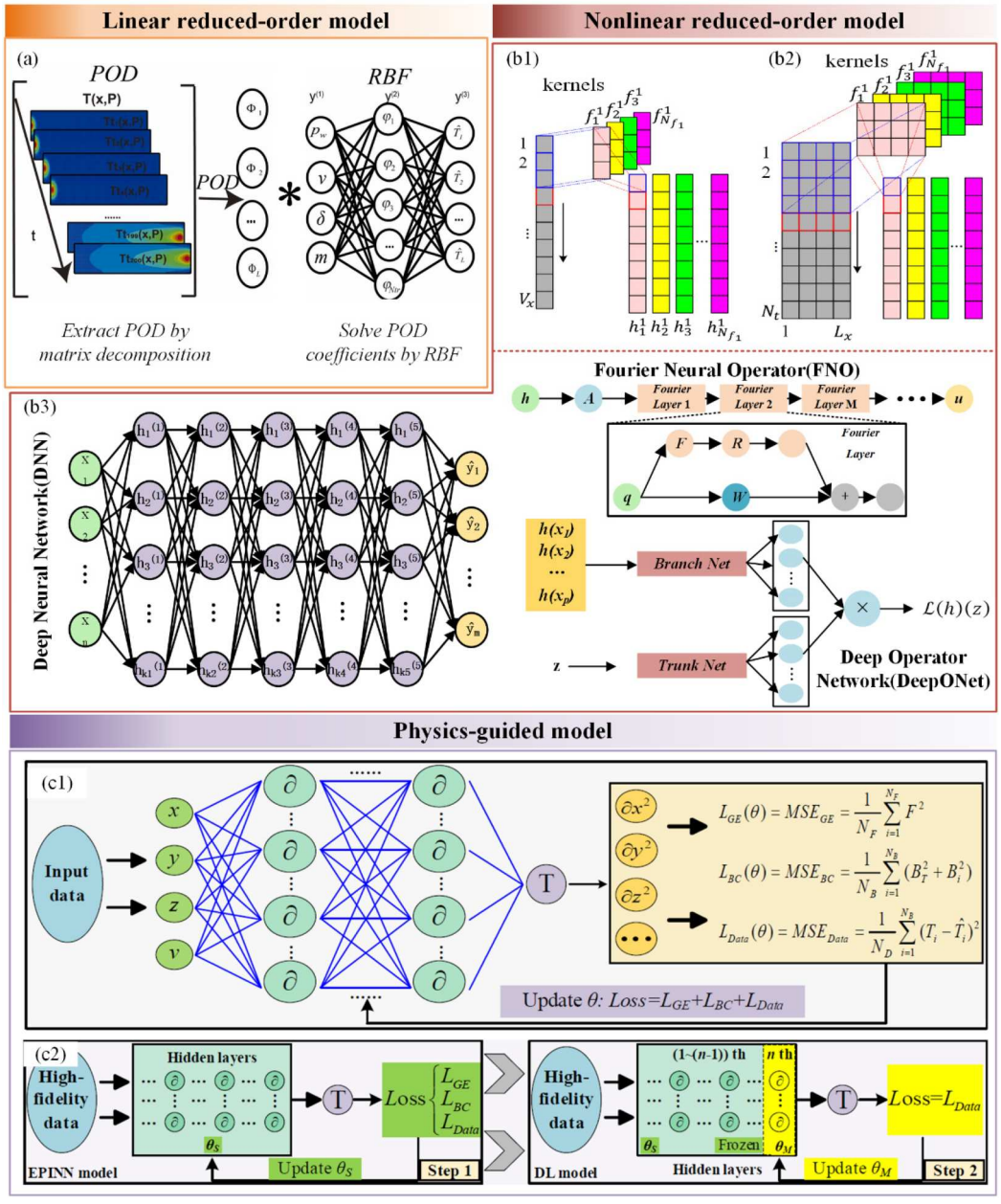


Figure 6. (a) Rapid prediction of the temperature field achieved through dimensionality reduction of multi-dimensional data [80]. (b1) and (b2) Spatial one-dimensional convolutional compression and temporal one-dimensional convolutional compression for strongly nonlinear problems [81]. (b3) Application of FNO and DeepONet to an additive manufacturing thermal model based on the MOOSE framework [82]. (c1) and (c2) Transfer learning enhanced physics-informed neural network (TLE-PINN) [83].

nonlinear ROMs based on deep neural networks. A comparative study evaluated a linear model (POD + artificial neural network) and a nonlinear model (convolutional autoencoder + multilayer perceptron) for melt pool temperature prediction. The nonlinear model achieved significantly higher accuracy owing to its nonlinear compression capability, and it also outperformed a conventional deep network using only cross-sectional data [84].

For flows with strong nonlinearity and discontinuities such as shock waves, a non-intrusive ROM framework based on a deep convolutional autoencoder was introduced. It employs a spatiotemporal dual compression strategy. A one-dimensional convolutional autoencoder first compresses the snapshot matrix spatially (Figure 6(b1)), after which a second 1D convolutional compression operates on the spatial latent vectors along the temporal dimension (Figure 6(b2)). This yields very low-dimensional spatiotemporal latent vectors, and a multilayer perceptron maps them to the random input parameters. Unlike conventional POD, which produces oscillations at discontinuities, this framework accurately captures shock features and provides smooth,

oscillation-free statistical moment predictions [81]. Beyond autoencoder-based approaches, neural operator methods such as the Fourier neural operator (FNO) and the deep operator network (DeepONet) offer another effective pathway. Applied to an additive manufacturing thermal model within the MOOSE framework (Figure 6(b3)), FNO and DeepONet directly learn the operator mapping from laser process parameters to the time series of melt pool temperature and volume [82]. These methods handle time-varying responses without resorting to linear dimensionality reduction and surpass traditional deep neural networks in both prediction accuracy and generalisation.

ROMs achieve rapid full-field prediction through linear or nonlinear compression. Linear POD-based methods are simple and efficient for smooth physical fields, while nonlinear methods using deep autoencoders capture strong nonlinearities and discontinuities with higher accuracy. The two can be selected or combined based on the problem's nonlinearity and real-time constraints, providing technical support for online monitoring of the PBF-LB process.

4.1.3. Physics-informed MODELS

Although statistical-learning models and many non-intrusive reduced-order models can rapidly predict process responses, their reliability often depends strongly on the representativeness of the training or snapshot data, and they may not explicitly enforce all governing physical constraints. Physics-informed models address this by embedding the governing partial differential equations (PDEs) and their boundary and initial conditions as residual terms in the loss function of a machine learning framework, so that the model satisfies physical constraints while fitting data. Depending on how physical knowledge is incorporated, they fall into two categories, namely physics-informed neural networks and physics-based feature engineering. The former directly uses PDE residuals as regularisation, while the latter extracts physically meaningful features from simulations or theoretical derivations to serve as model inputs. To highlight the distinctive characteristics of physics-informed models, Table 4 summarises the distinctions and overlaps among physics-informed models, statistical-learning-based surrogate models, and conventional mechanistic models.

Physics-informed neural networks (PINNs) are the most representative physics-informed models. As shown in Figure 6(c1), they consist of a deep neural network and a loss function that includes residuals of the governing PDEs together with initial and boundary conditions. Taking the PBF-LB thermal analysis as an example, the network takes spatial coordinates, time, process parameters, and other variables as input and outputs the temperature field. Automatic differentiation provides the output derivatives needed to include the PDE residuals in the loss function [85, 86]. This formulation can produce physically constrained solutions with limited labelled data, although the result still depends on the governing equations, boundary conditions, and optimisation procedure.

Physics-informed models offer distinct advantages over purely data-driven approaches. First, their predictions inherently satisfy fundamental physical laws such as energy conservation, improving interpretability and avoiding non-physical results that black-box models may produce. Second, they exhibit higher data efficiency, allowing PINNs to be trained and to deliver parametric predictions with sparse experimental data [86]. Third, PINNs can provide continuous field solutions and their gradients with respect to input parameters, greatly facilitating inverse calibration and sensitivity analysis, for instance in the calibration of heat source model parameters [85].

In PBF-LB, physics-informed models have been applied to temperature-field prediction and heat-

Table 4. Boundaries and overlaps among mechanistic, statistical-learning-based surrogate, and physics-informed models.

Modelling approach	Core definition	Main inputs	Typical outputs / roles	Domain limits and validation requirements
Conventional mechanistic models	Solves discretized conservation equations with constitutive relations and boundary conditions	Physical laws, material properties, initial/ boundary conditions, and calibration data	Temperature, flow, stress, phase, interface, and microstructure fields; mechanism testing	Valid only under stated constitutive assumptions; validate fields and integral quantities independently
Statistical-learning-based surrogate models	Fits low-dimensional input-output mappings using response surfaces, Gaussian processes, or related regression methods	Designed experimental or high-fidelity simulation samples; optional dimensionless descriptors	Rapid scalar prediction, process-window mapping, optimisation, sensitivity analysis, and uncertainty estimation	Mainly interpolative; use space-filling sampling, grouped validation, uncertainty checks, and out-of-domain tests
Physics-informed models	Embeds governing-equation residuals, conservation, initial/ boundary conditions, or other physics in a learned approximation	Collocation points or limited data plus explicit physical laws and stated enforcement mechanisms	Parameterized forward/inverse solutions, parameter identification, and physically regularised prediction	Physics constraints do not ensure accuracy; report residuals, data-fit error, loss weighting, ablation studies, and external validation

source calibration. A PINN-based thermal model can predict transient temperature fields and melt pool dimensions for arbitrary process parameter combinations at much lower computational cost, with accuracy comparable to finite element benchmarks [87]. The transfer-learning-enhanced PINN (TLE-PINN) (Figure 6(c1,c2)) uses a two-stage training procedure and locally adaptive activation functions to improve prediction accuracy and convergence when high-fidelity data are available [83]. Coupling a PINN with a cellular automaton produces an integrated thermal-microstructure simulation framework. The efficient gradient computation of PINNs enables inverse calibration of the heat source, and the calibrated temperature field is then fed into the CA model for microstructure prediction [85]. Physics-based feature engineering has also been used. By extracting a few physically intuitive features from melt pool images (e.g. melt pool length, temperature distribution, and ejecta characteristics) and feeding them into a simple machine learning model, high-accuracy porosity classification can be achieved, matching the performance of deep learning models that directly use the images [86]. This result indicates that physically motivated features can reduce the need for complex architectures and large training datasets.

Despite their promise, physics-informed models still face several challenges. PINN training is highly sensitive to the distribution of collocation points. Near the laser heat source, where steep temperature gradients exist, conventional sampling strategies often cause training failure, necessitating physically inspired adaptive collocation strategies [87]. PINNs also encounter substantially increased training difficulty when handling strong nonlinearities such as latent heat and temperature-dependent material properties, and they are difficult to directly extend to multitrack, multilayer, or part-scale simulations. For physics-based feature engineering, the feature design lacks universality, as the feature set may need to be redefined for different material or process conditions.

In summary, physics-informed models integrate physical laws with data-driven methods by embedding PDE residuals into the loss function or by extracting physically intuitive features. PINNs offer unique advantages in efficiency and differentiability for parametric temperature field prediction and inverse calibration, whereas physics-based feature engineering holds practical value in edge monitoring scenarios with limited computational resources. Such models are still at an early stage and are expected to be extended toward multiphysics coupling, cross-scale modelling, and more complex geometries, serving as an important bridge connecting mechanistic models and data-driven methods.

4.2. Machine learning

Physics-informed models improve interpretability and data efficiency through PDE constraints, yet they still require explicit physical modelling. When physical models are incomplete or computationally prohibitive, purely data-driven machine learning provides an alternative by automatically extracting defect-related features from massive high-dimensional monitoring data and establishing end-to-end mappings from process signals to quality attributes. This section focuses on two core tasks, defect diagnosis and quality property prediction, with an emphasis on recent advances in deep learning and advanced regression methods.

Convolutional neural networks (CNNs) are widely used for defect diagnosis because they extract spatial features directly from images. A lightweight CNN designed for thermal imaging directly identifies delamination and spatter, achieving 96.80% accuracy, and Grad-CAM heatmaps confirm the network's focus on abnormal temperature regions [88]. For cracks, which are difficult to detect optically, acoustic emission (AE) signals combined with machine learning provide information that is not available from optical images. Features such as spectral principal components or statistical moments fed into classifiers like extreme gradient boosting or random forest can yield crack identification accuracy up–99% [89].

Melt pool image analysis employs increasingly diverse strategies. To resolve the localisation difficulty caused by the lack of synchronisation with the laser position, a self-supervised learning method with a Kalman filter was proposed to predict scanning direction and recursively estimate melt pool position, establishing spatial references for anomaly diagnosis [90]. For higher detection sensitivity, high-speed melt pool imaging with a gradient boosting decision tree model enabled voxel-level porosity classification [91]. By designing the part geometry to induce natural porosity and adopting the 90/95 probability of detection criterion from non-destructive evaluation, the model met the required standard at a porosity level of 0.11%, with feature analysis highlighting melt pool shape and spatter behaviour as the most critical diagnostic indicators. To address cross-material generalisation, a variational autoencoder was employed to extract latent space features from AE signals. Without retraining, these features could correctly discriminate quality level trends for different materials, offering a route toward material-agnostic diagnostic models [92].

Quality property prediction shifts the focus from discrete defect classification to the continuous regression

of mechanical properties and porosity. A multi-scale CNN was used to detect six types of powder bed anomalies from visible-light post-recoating images, including recoater hopping, streaking, debris, bumps, part damage, and incomplete spreading [88]. By feeding local, mesoscale, and global image patches into the three colour channels of a pretrained network, the model learns fine-grained features and contextual information simultaneously, outputting pseudo-probabilities through a softmax layer with an overall classification accuracy of 97% [88].

For continuous quality indicators, Gaussian process regression (GPR) has been applied to establish a quantitative relationship between process parameters and porosity. With laser power and scan speed as inputs, GPR estimates model parameters through Bayesian inference and provides both point predictions and uncertainty estimates of porosity, achieving accurate predictions from only 42 initial experimental samples and further reducing error through adaptive sampling [89]. GPR reveals the physical influence patterns of power and speed on porosity, and its prediction confidence intervals support process window optimisation and experimental cost control. Physics-informed neural networks and Bayesian methods have also been introduced to enhance model interpretability and robustness [93, 94].

Data leakage is a particular concern in PBF-LB because images and signal windows that appear to be separate samples often come from the same track, layer, coupon, or build. Random frame-level splitting can place nearly identical neighbouring frames, overlapping temporal windows, or pixels from the same physical defect in both the training and test sets. Performing data augmentation before splitting creates a similar problem. Leakage can also occur when normalisation, feature selection, dimensionality reduction, class balancing, or hyperparameter selection uses information from the full dataset. These practices inflate the apparent accuracy and underestimate uncertainty because the test set is no longer independent [95, 96].

A valid evaluation must define the physical unit of independence before the data are split. Frames, windows, voxels, and augmented variants from the same track, layer, coupon, or build should remain in one group. Grouped or blocked cross-validation by build is appropriate for evaluating performance within a machine [97]. Tests that exclude one region of the parameter space assess interpolation and extrapolation, while tests that exclude one material, machine, geometry, or build condition assess domain transfer. Preprocessing and feature selection must be fitted within each training fold. Model selection should use nested cross-validation or a separate validation set, and the final test set should remain unused until the model and decision thresholds have been fixed [98, 99]. If only one build is available, the results support internal consistency but not generalisation to the process as a whole [96].

Reports should distinguish performance at the sample level from performance at the build level and should state the number of independent builds, coupons, layers, and defects [96]. Accuracy alone is not sufficient for an imbalanced detection problem. More informative measures include precision, recall, F1 score, the area under the precision-recall curve, false alarms per layer, probability of detection as a function of defect size, and localisation measures such as intersection over union or mean average precision [100]. Regression and forecasting errors should be reported for each build and operating regime, together with calibration or prediction intervals, forecast horizon, and inference latency [101, 102]. External validation should document changes in sensor gain, optics, material, powder lot, geometry, machine, and gas flow [102, 103]. Domain adaptation or transfer learning should be compared with a model trained without adaptation and evaluated on target-domain data that were not used to tune the adaptation procedure [12, 103].

Machine learning has progressed from shallow classification to convolutional localisation, self-supervised representation learning, probabilistic regression, and time-series forecasting. Evaluation should now focus on physically independent test data, calibrated uncertainty, robustness to class imbalance, useful prediction lead time, and transfer across builds and operating domains rather than on a single accuracy value obtained from a random split. Physics-guided features and auxiliary tasks may improve data efficiency and interpretability, but they do not remove the need for independent validation. Models intended for closed-loop control should also report the total latency of sensing, preprocessing, inference, and actuation.

5. Data-mechanism dual driven analysis

Building on the monitoring and modelling methods discussed above, this section traces how melt pool dynamics, vapour-plume behaviour, spatter, and denudation produce gas porosity, lack of fusion, cracking, and segregation. The aim is to move beyond statistical association and establish a physically interpretable causal chain.

5.1. Melt pool dynamics

The melt pool, where energy input and material transformation occur, features fluid flow that directly

governs heat distribution, solute transport, and solidification behaviour. As the starting point of the entire physical process, melt pool dynamics establish the hydrodynamic foundation for subsequent vapour generation and spatter formation.

5.1.1. Marangoni convection

In the PBF-LB process, Marangoni convection, a thermocapillary flow driven by surface tension gradients, exerts a decisive influence on melt pool evolution, heat transport, and solidification structure. The combination of particle image velocimetry (PIV) and synchrotron X-ray imaging reveals that the velocity distribution of Marangoni convection is not a simple single-peak or double-peak pattern but a multi-vortex structure (Figure 7(a1)). A clockwise vortex exists at the front of the melt pool and a counterclockwise vortex at the rear [109]. This double-vortex structure is the main cause of solute enrichment and bubble entrapment within the melt pool.

The driving force originates from the surface tension difference induced by the temperature gradient on the free surface. Surface tension is lower in the high-temperature region and higher in the low-temperature

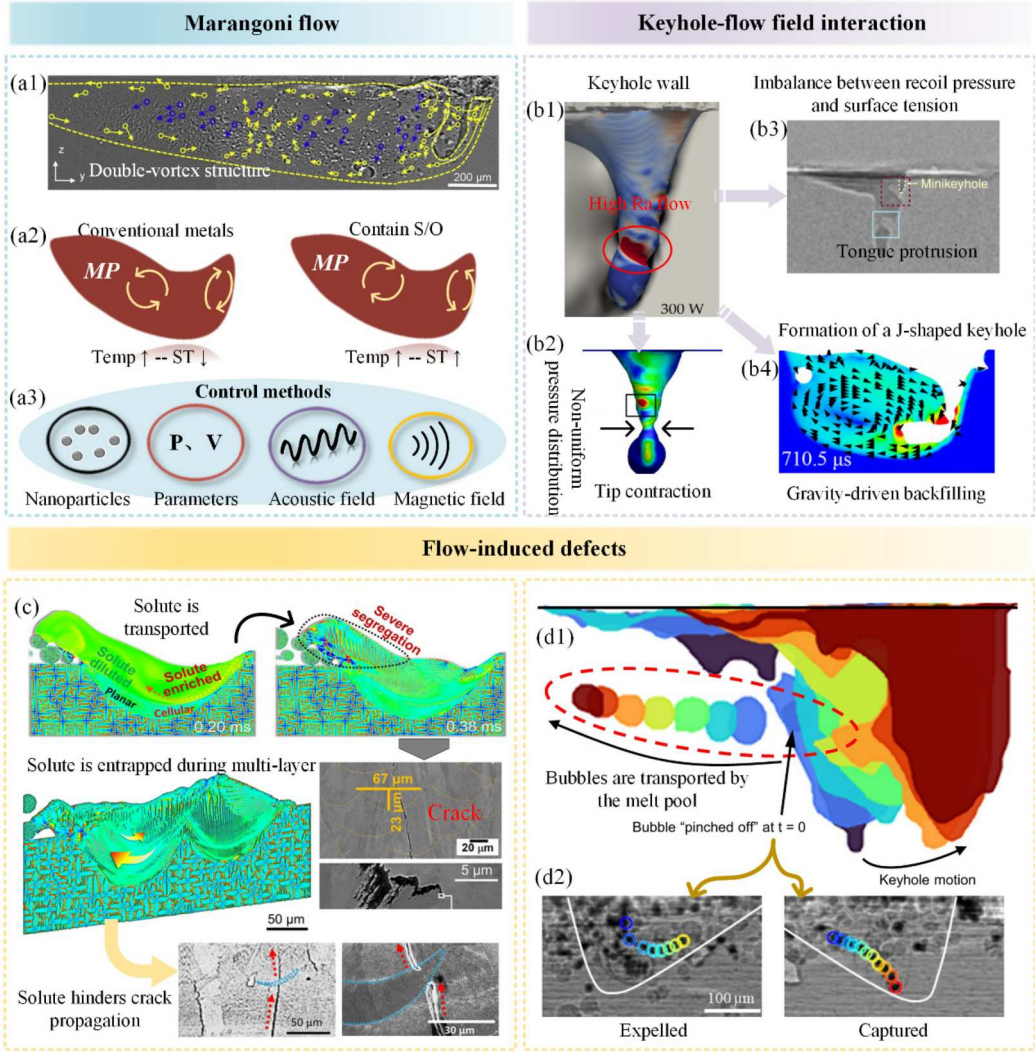


Figure 7. (a1) Double-vortex Marangoni convection observed under X-ray imaging [104]. (a2) When surface-active elements such as sulfur or oxygen are present in the melt, the temperature coefficient of surface tension changes from positive to negative at a critical temperature, generating two Marangoni flows in opposite directions. (a3) Strategies for controlling Marangoni convection. (b1) High-Reynolds-number melt flow on the keyhole wall [5]. (b2) Complementary protrusion pairs formed on the front and rear keyhole walls due to melt pool fluctuations, leading to keyhole collapse [105]. (b3) Periodic wavy structures on the front keyhole wall, some exhibiting tongue-like shapes [106]. (b4) Melt backfilling disrupts the keyhole structure [105]. (c) Melt pool flow affects solute distribution at the solidification front, thereby regulating segregation, and under multilayer build conditions, solute trapping effects interrupt crack propagation [107]. (d1) Bubbles detach from the keyhole and are transported to the rear of the melt pool [108]. (d2) Two outcomes of bubble movement in the melt pool are escape and entrapment [89].

region, which drives the melt from the centre to the edge [110, 111]. However, surface-active elements such as sulfur or oxygen can change the temperature coefficient of surface tension from positive to negative above a critical temperature. Consequently, the maximum surface tension appears along a certain isotherm, generating two opposite Marangoni flows (Figure 7(a2)). These flows meet in a stagnation zone and produce a downward jet, which is an important source of flow instability and turbulence [112]. Melt pool geometry also significantly affects the spatial structure of Marangoni convection. In a ring-shaped shallow pool, Marangoni convection excites circumferentially propagating thermocapillary waves, and the wave number decreases as the ring gap increases. An applied axial magnetic field can effectively suppress non-axisymmetric flow, making the isotherms concentric and thereby achieving stable control of the convection [109].

In addition to applied energy fields, process parameters and the addition of reinforcing phases also regulate the intensity and mode of Marangoni convection (Figure 7(a3)). Higher laser power and lower scan speed increase the temperature gradient and enhance Marangoni convection, resulting in significantly greater melt pool width and depth, whereas low power and high speed reduce the melt pool size and increase the cooling rate [112]. In ring laser processing of nanocomposites, Marangoni convection dominates the formation of multiple vortex rings, with melt flowing from the centre to the edge and forming a heat accumulation zone at the rear. The addition of nanoparticles suppresses convection intensity by increasing the effective viscosity of the melt. Moreover, interfacial thermal resistance induces a heat retention effect that expands the molten zone and shrinks the heat-affected zone [113]. Nanoparticles also experience thermophoretic forces that are enhanced in regions of large temperature gradients, driving them from high-temperature to low-temperature zones and in turn influencing the local Marangoni convection distribution. These results show the complex behaviour of Marangoni convection under various material systems and process conditions. The coupling among the temperature field, physical property parameters, and applied fields is crucial for understanding melt pool dynamics and achieving precise process control.

5.1.2. Complex flow field and keyhole interaction

A strong coupling between the complex flow field in the melt pool and the keyhole is critical for understanding keyhole instability and gas pore generation. The melt pool flow state is directly influenced by laser process parameters, which in turn determine the keyhole dynamics. Under high laser power, localised intermittent high-Reynolds-number regions appear in the melt pool, indicating that the flow is entering a laminar-to-turbulent transitional regime [114]. As shown in Figure 7(b1), this high-Reynolds-number flow generates complex flow structures and induces a non-uniform dynamic pressure distribution on the keyhole surface, where alternating positive and negative pressures act on the wall. This pressure interface tears the keyhole tip apart and triggers gas pore formation. With a powder bed, the non-uniformity of the powder layer thickness disrupts the force balance at the keyhole rim, serving as a major external factor that triggers fluctuations in the keyhole depth [5].

During keyhole fluctuation, complementary protrusion pairs form on the front and rear walls (Figure 7(b2)). Owing to the periodic competition between recoil pressure and surface tension, ripple structures appear on the front keyhole wall (Figure 7(b3)), and their wavelength decreases with increasing scan speed [115]. The evolution frequency of these ripples closely matches that of subsequent gas pore generation, indicating that front wall ripples are precursors to pore formation. Rear wall protrusions primarily arise from focused heating by the laser beam reflected from the front wall. Driven by recoil pressure, the front and rear protrusions move toward each other and eventually merge, causing the keyhole to pinch off and trap gas as bubbles inside the melt pool [105]. If the bubbles fail to escape before solidification, they become porosity defects.

In addition to pinching, the keyhole can also collapse through backfilling. As shown in Figure 7(b4), when the keyhole bottom bends into a J-shape and the laser can no longer effectively heat it, the recoil pressure weakens, surface tension becomes dominant, and the melt fills the keyhole from the bottom upward [105]. Unlike pinching, gas can usually escape from the top during backfilling, making pore formation less likely, though it can cause substantial fluctuations in the keyhole depth. Quantitative force analysis further reveals the coupling between the flow field and the keyhole. Recoil pressure is the dominant force driving expansion of the front keyhole wall, at the micronewton level, while surface tension predominantly compresses the rear keyhole wall. The dynamic balance between these two forces at the keyhole bottom governs the keyhole depth [5]. Although the Marangoni force acts over the entire melt pool interface, it is an order of magnitude smaller than the former two and has limited direct influence on keyhole evolution. In the melt pool tail far from the keyhole, however, surface tension and

the Marangoni force dominate, generating a centrifugally directed vortex flow.

5.1.3. Flow-induced defect formation

Melt pool flow is the core physical mechanism behind various metallurgical defects. Solute transport is directly governed by melt convection. When the flow direction opposes the solute concentration gradient at the solidification front, solutes accumulate locally, increasing constitutional undercooling and promoting the formation of inter-dendritic channels. Such channel segregation, known as freckles, constitutes a typical segregation defect in directional solidification and additive manufacturing [116]. The geometric sensitivity of the flow pattern further aggravates this problem. In regions with abrupt cross-sectional change at the solidification front of the melt pool tail, the shrinkage-induced feeding flow is amplified, creating strong negative segregation zones. In expanding regions at the solidification front of the melt pool front, the altered flow pattern increases the risk of stray grain formation [116]. As shown in Figure 7(c), the melt pool flow directly determines the severity of segregation by influencing the solute distribution at the solidification front [107].

Gas pore formation is closely related to melt pool flow, with unstable keyhole collapse as the primary source. The interaction between the recirculating vortex at the rear of the melt pool and the high-speed forward convection creates protrusions on the rear keyhole wall, obstructing laser energy absorption and triggering periodic keyhole collapse. As shown in Figure 7(d1, d2), the resulting bubbles are subsequently entrained by the flow into the tail of the melt pool. If the solidification rate exceeds the bubble rise velocity, they are trapped by the solid-liquid interface and form gas pores [89]. Bubbles exhibit a spiral ascent pattern, and their rise velocity depends on the melt pool flow. Bubbles at vortex centres ascend most slowly and are easily captured, whereas those at the periphery rise faster and have a markedly higher escape probability [117]. Circulating flow and vortices can also sweep bubbles deeper, while low surface tension and high viscosity further impede their rise and escape [118].

Hot crack formation depends on the liquid feeding capacity in the mushy zone, which is controlled by melt pool flow through the permeability of inter-dendritic channels. When the flow aligns with the heat flow direction, solute enrichment lowers the solidus and extends the solidification interval. If inter-dendritic liquid flow cannot fill the shrinkage voids, cracks form under thermal stress [107, 118]. In highly segregating nickel-based superalloys such as CM247LC, strong hafnium enrichment creates low-melting eutectic phases in inter-dendritic regions. These phases undergo liquation during subsequent thermal cycles, severely impairing printability. However, tailoring process parameters to promote solute trapping can produce a fine cellular microstructure at the melt pool bottom, as shown in Figure 7(c). The resulting uniform solute distribution significantly reduces local crack susceptibility, effectively suppressing crack propagation even in difficult-to-print alloys [107]. Thus, actively controlling melt pool flow to optimise solute distribution is key to suppressing hot cracks. Moreover, exploiting flow mechanisms to introduce reinforcing phases can also suppress cracks. Marangoni convection transports reinforcing particles to inter-dendritic regions, refining grains and hindering crack propagation, thereby improving both room-temperature and high-temperature strength [119].

Melt pool flow affects segregation, gas pores, cracking, and surface defects through solute transport, bubble migration, liquid feeding, and interfacial stability. Quantifying these relationships is necessary for process optimisation, defect mitigation, and the design of active flow-control methods such as magnetic-field assistance.

5.2. Evaporation and plume dynamics

Strong melt pool flow and keyhole oscillations affect solidification and gas-pore formation and also drive metal evaporation. When the temperature exceeds the boiling point, vigorous vaporisation generates the recoil pressure that drives keyhole formation and melt pool flow. The resulting vapour plume is highly dynamic and affects energy transfer, powder motion, and build quality.

5.2.1. Vapour and recoil pressure

When a high-energy laser beam interacts with a metallic material, the local temperature rapidly rises above the boiling point, inducing intense evaporation and a high-velocity metal vapour jet (Figure 8(a1, a2)) [71, 108, 120, 124-126]. The vapour generation rate is jointly governed by the thermophysical properties of the material and the laser energy input. Low-boiling-point elements such as zinc and magnesium evaporate preferentially in alloys, causing a significant compositional difference between the vapour and the original powder [71, 124, 126]. In aluminum alloys, the high activity of magnesium leads to notable evaporation, making the magnesium content in the vapour much higher than in the original powder [71]. In Cu-10Zn alloy, the boiling point of zinc (1180 K) is far below that of copper, so zinc evaporates preferentially and

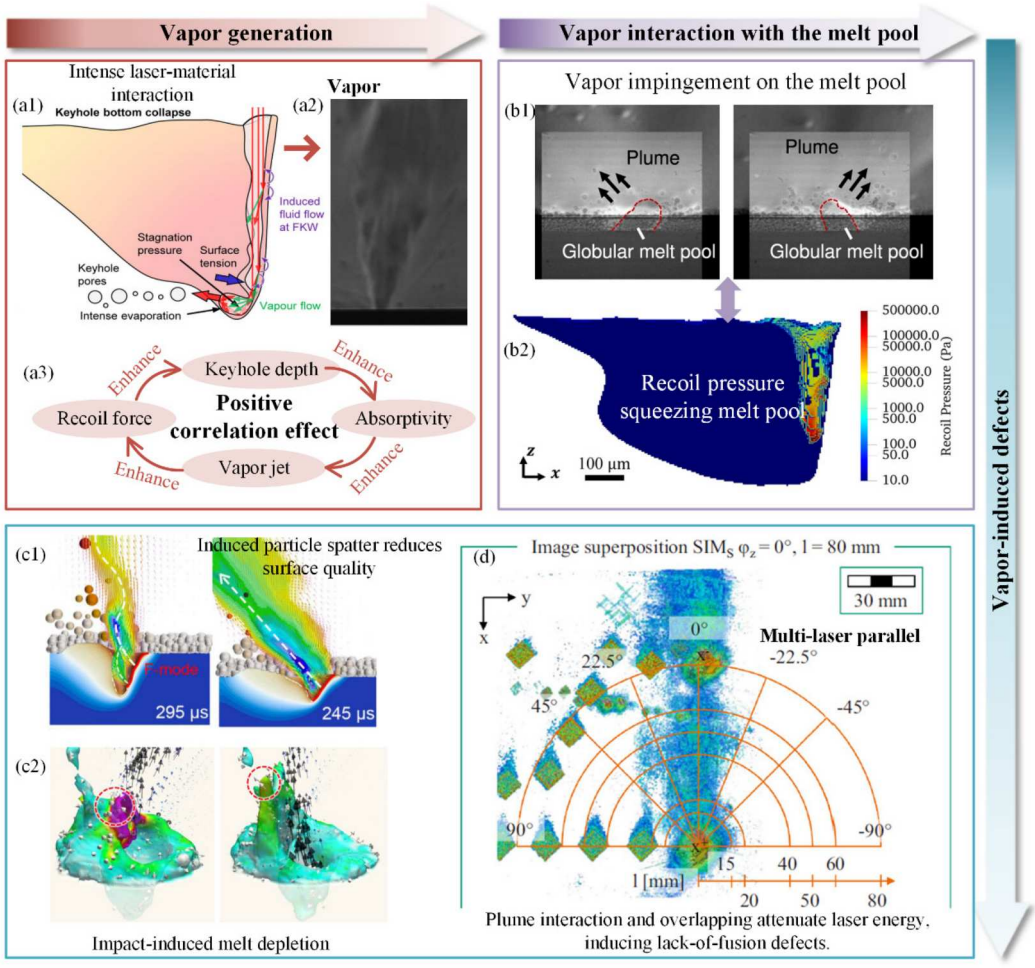


Figure 8. (a1) Enhanced recoil pressure deepens the keyhole, and the increased keyhole depth in turn intensifies laser reflections within the keyhole [108]. (a2) Vapour jetting phenomenon observed under schlieren imaging. (a3) Positive feedback phenomenon in vapour jetting. (b1) Vapour impingement on the melt pool causing melt protrusions [34]. (b2) The region of action of vapour recoil pressure on the melt pool surface [120]. (c1) Variation of vapour plume angle with process parameters, where the plume impacts the powder bed and generates particle spatter [121]. (c2) Backward-tilted vapour jet impinging on the melt pool and inducing droplet spatter [122]. (d) Coverage area and intensity of vapour plumes in multi-laser configurations [123].

substantially. Its saturated vapour pressure is one to four orders of magnitude higher than that of copper within the melt pool temperature range [124]. The evaporation rate is often described by the Langmuir model, in which the flux scales with the saturated vapour pressure and depends on temperature and molar mass [124, 126]. In the actual PBF-LB process, the Knudsen layer forming at the melt pool surface suppresses continuous evaporation, and a portion of the metal vapour can re-condense and fall back into the melt pool. Consequently, the net evaporation rate is lower than the theoretical maximum [120, 124–126].

The escaping vapour exerts a downward recoil pressure on the melt pool surface (Figure 8(b2)), which is the primary driving force for keyhole formation and sustenance. Its magnitude is typically expressed as a function of atmospheric pressure, latent heat of evaporation, and temperature, and increases exponentially with melt pool temperature [108, 120, 124, 126]. Recoil pressure drives the liquid metal outward and downward from the laser interaction centre, significantly altering the melt pool morphology [108, 120, 124-126]. Furthermore, the vapour possesses high kinetic energy. As it exits the keyhole, it transfers a portion of this energy to the melt pool, causing violent oscillations (Figure 8(b1)) and inducing droplet spatter.

Investigations combining multiphysics simulation and synchrotron radiation validation reveal that recoil pressure is not uniform on the keyhole wall. It concentrates in localised hot spots where the temperature exceeds the boiling point and migrates dynamically with laser scanning. The resulting periodic, asymmetric loading contributes to keyhole instability [7, 108]. As shown in Figure 8(a3), stronger recoil pressure

deepens the keyhole, which in turn increases laser absorptivity through multi-reflection, creating a positive feedback loop [108, 120, 125, 126]. Thermo-fluid simulations confirm that this non-uniform loading is a key factor in keyhole instability and collapse [120]. When the balance among recoil pressure, surface tension, and hydrodynamic pressure is disrupted, the rear keyhole wall collapses periodically, forming transient bubbles [108, 120, 124]. In materials with highly volatile elements such as Cu-10Zn alloy, a sharp rise in vapour pressure can even trigger micro-explosions of the melt pool [124]. These micro-explosions occur because the melt pool temperature can spike by hundreds to thousands of Kelvin in extremely short timescales, causing the saturated vapour pressure of zinc to surge to tens or even hundreds of atmospheres, far surpassing the melt surface tension [124]. The resulting impact can cause melt spatter and create crater-like defects in the scan track, severely compromising the build quality.

5.2.2. Influence of vapour behaviour on build quality

The vapour plume behaviour and its instability are directly correlated with typical defects such as gas pores, spatter, and lack of fusion. Characterising plume evolution and its relation to process stability is therefore relevant to quality control. As shown in Figure 8(c1), the vapour plume evolution occurs in two stages. Upon laser initiation, a transient stage occurs where the vapour jet ejects vertically upward at over 100 m/s, and this stage is highly similar across different parameters [121]. As scanning proceeds, the vapour behaviour becomes parameter-dependent and the ejection direction inclines. The vapour jet angle correlates positively with scan speed, transitioning from an acute to an obtuse angle at higher speeds [121, 127]. An inclined jet increases the contact area with the melt pool, intensifying melt disturbance and raising the probability of droplet spatter (Figure 8(c2)) [122].

High-speed camera monitoring reveals that the plume morphology, direction, and intensity show typical transitions with melting mode. In conduction mode the jet is vertical, in transition mode it tilts backward, and in keyhole mode the direction becomes chaotic. This transition reflects changes in the melt pool front wall inclination and melt flow state [128]. Combined schlieren and synchrotron X-ray imaging quantitatively establishes a linear relationship between the vapour jet angle and the keyhole front wall angle, and shows that the vapour jet angle change lags behind that of the front wall angle. This hysteresis characterises the coupling dynamics between the keyhole and the vapour plume [108].

Vapour plume instability is essentially an oscillation originating from the dynamic shift of the localised enhanced recoil pressure position on the keyhole wall [121]. A quantitative correlation links the frequency characteristics of vapour oscillations to defect size. Spectral analysis of acoustic emission (AE) signals acquired by high-frequency microphone arrays or acoustic sensors can reveal vapour and liquid-gas interface oscillations within the keyhole. In the conduction mode, AE signals mainly appear in the lower frequency band (approximately 5–15 kHz). In the keyhole mode, higher-frequency components (on the order of tens of kHz) become prominent, associated with liquid-gas interface oscillations and vapour jet instability. These frequency components reflect the coupling between vapour oscillations and melt pool dynamics [36], providing a physical basis for inferring defect types from acoustic signals. Furthermore, vapour oscillations can induce the shielding gas to form a counter-rotating double-vortex structure, causing periodic variations in the drag force on powder particles and explaining the generation of backward and lateral spatter.

In multi-laser parallel processing, the vapour plume also exhibits spatial oscillations (Figure 8(d)). The plume from one melt pool diffuses along the gas flow, forming a vapour zone. If another laser beam initiates or passes through this zone, it interacts with nanoparticles in the vapour, and the newly generated vapour oscillates unstably. This attenuates the laser energy and degrades beam quality, leading to melt pool instability and lack-of-fusion defects [123].

Overall, the vapour plume behaviour is coupled with scan speed, melting mode, melt flow, build position, and multi-laser coordination. The oscillation characteristics and spatial distribution of the plume can serve as important indicators of process stability. Optimising process parameters to maintain stable periodic oscillations and selecting an appropriate scan path can reduce gas pores and spatter, thereby improving build quality [121, 127]. In addition, in situ monitoring of vapour plume behaviour may provide a signal for real-time control and process window optimisation [128].

5.3. Spatter behaviour

Spatter in the PBF-LB process refers to the ejection of molten material and solid particles caused by intense metal vaporisation and melt flow under extremely high laser energy density. As an inherent byproduct of the multiphysics coupling among metal vapour recoil pressure, keyhole instability, and shielding gas flow, spatter can be classified by material source and morphology. Its complex dynamics and trajectories directly

affect process stability. Moreover, the generation, transport, and deposition of spatter particles are significant sources of internal defects, powder bed contamination, and deterioration of mechanical properties. An in-depth understanding of spatter generation mechanisms, motion characteristics, and detrimental effects is therefore essential for optimising process parameters, suppressing defects, and improving the build quality of PBF-LB.

5.3.1. Spatter types and formation mechanisms

Spatter can be broadly classified into solid spatter and liquid spatter based on material form and generation mechanism. Molten droplet spatter originates from melt pool instability, and its formation mechanisms mainly include the following types. The first is the recoil-pressure-dominated mechanism, in which the high-pressure vapour jet from violent keyhole vaporisation ejects melt pool liquid at high velocity, forming spherical or near-spherical droplets with high speed and relatively regular shape. The second is the synergistic mechanism of the Marangoni effect and recoil pressure, where surface tension flow induced by the temperature gradient transports melt to the pool rim and the vapour pressure subsequently expels it. This type often has unmelted powder particles adhering to it. The third is the thermal-effect and shock-wave-dominated mechanism, where the shock wave generated by the laser on the melt pool front or already solidified region can splash melt or semi-solid material forward, forming irregular blocky agglomerates with short flight distance [136].

High-speed imaging has further revealed more specific spatter categories, such as metal jet spatter formed by breakup of liquid protrusions at the edge of the melt pool depression, powder agglomeration spatter formed by droplets adhering to powder or droplet merging, entrained melt spatter formed when powder is carried into the heat source zone by gas flow, melted, and then ejected, and defect-induced spatter triggered by pre-existing defects in the layer (Figure 9(a,b)) [129]. These types differ notably in velocity, size, and morphology. For instance, the average velocity of metal jet spatter can reach approximately 2.17 m/s, while powder agglomeration spatter tends to have a larger average size but lower velocity.

From the perspective of multiphase interactions, the core process of spatter formation is laser-induced metal evaporation, which produces a high-velocity vapour jet. This jet, together with the gas vortices it induces, constitutes the dominant gas-phase flow field driving spatter [137]. Solid or unmelted powder spatter mainly arises from entrainment and accelerated ejection of powder particles by this flow field [137]. Molten droplet spatter, in contrast, is directly caused by vapour recoil pressure or vapour impact on the melt pool liquid [136, 137]. Spatter formation is thus inherently a multi-stage cascading dynamic process across the melt pool, gas phase, and powder phase.

Spatter characteristics also depend on the processing conditions. The velocity and direction of the shielding gas affect the melt pool surface and the vapour plume, thereby changing the number, size, and ejection angle of spatter particles. Higher gas velocity may increase the number of fine particles [72]. Scan direction is also important. Scanning against the gas flow generally reduces the amount of spatter and narrows its dispersion [72, 73]. The powder bed on one side of the melt pool and consolidated material on the other create an asymmetric thermal field. Spatter is therefore ejected mainly from the consolidated side, where heat is conducted more rapidly [73]. Turns and other discontinuities in the scan path can produce transient melt pool instability and a local increase in spatter [73]. These effects show why spatter measurements should be interpreted in relation to the gas flow, scan direction, and local build geometry.

5.3.2. Spatter dynamics

Spatter dynamics begin with the driving force acquired upon detachment from the melt pool or powder bed. This force originates mainly from the metal vapour jet generated by laser-material interaction and the induced shielding gas vortex (Figure 9(c1,c2)). Because spatter formation involves highly coupled gas, melt, and powder flows, no single experimental method can simultaneously capture the full multiphase behaviour. Coupled CFD-DEM models combined with high-speed imaging are therefore employed to resolve the generation, acceleration, and transport mechanisms of spatter [71, 74, 138].

In these models, the gas flow is solved by the Navier-Stokes equations and vapour fluxes by an evaporation model, while powder motion is described by the discrete element method (DEM) including fluid drag, wet adhesion, and inter-particle contact forces. This approach reproduces the vapour-jet-driven initial acceleration. The jet intensity depends strongly on alloy composition, particularly the volatility of low-boiling-point elements [71]. For example, AlMg5 alloy containing highly active magnesium produces a violent jet exceeding 160 m/s and a broad influence, whereas the less volatile AlMn5 alloy generates far weaker spatter. Quantitative studies show that spatter initial velocities lie mainly in the range 75–200 m/s (average approximately 129 m/s), and the upward ejection angle is concentrated

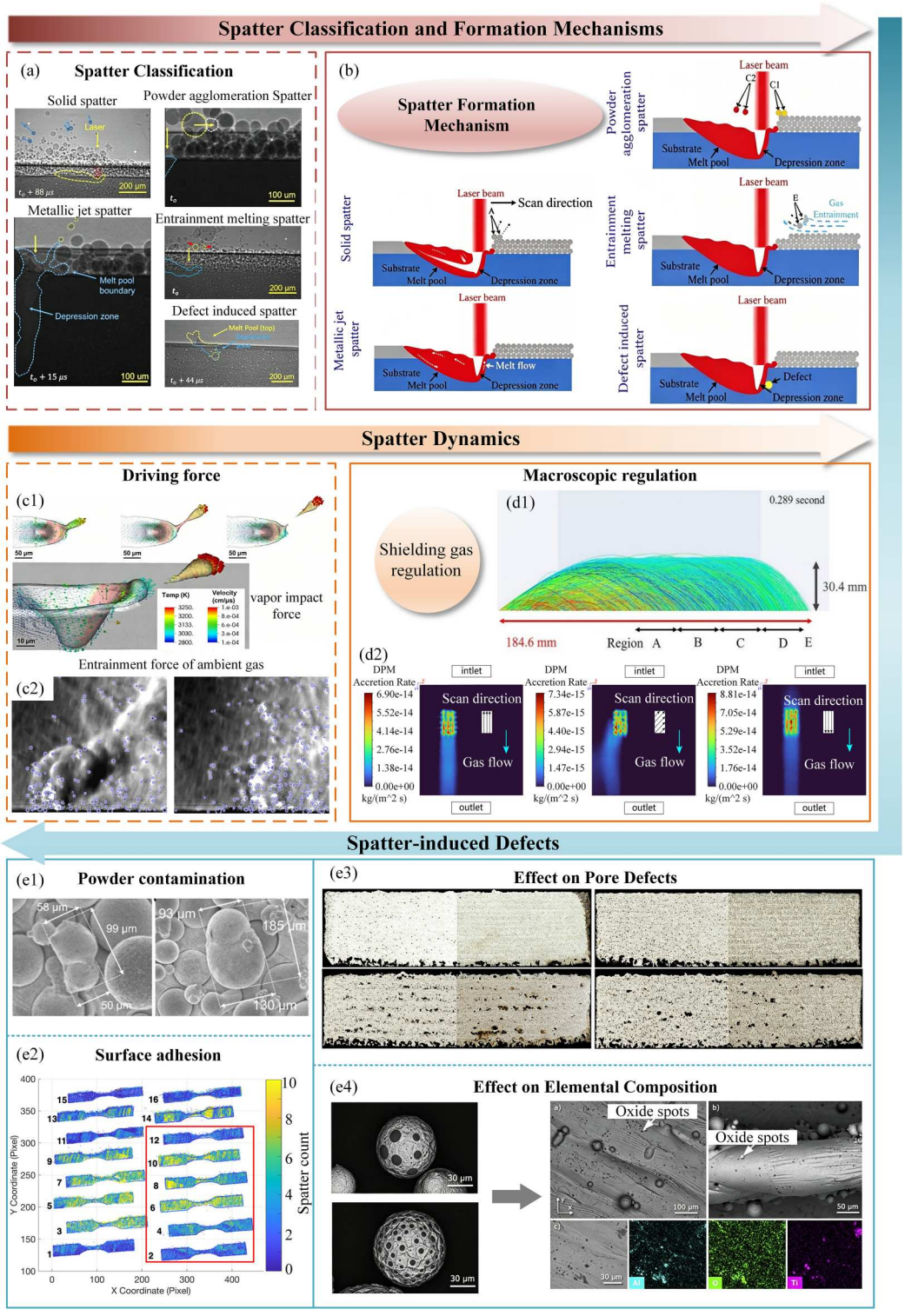


Figure 9. (a) Five spatter types, namely solid spatter, metal jet spatter, powder agglomeration spatter, entrained melt spatter, and defect-induced spatter [129]. (b) Schematic of the formation mechanisms of the five spatter types [129]. (c1) Spatter generation induced by the impact force of the vapour jet [130]. (c2) A large amount of solid spatter caused by the entrainment force of the ambient gas [33]. (d1) Removal effect of the shielding gas on spatter particles of different diameters [131]. (d2) Influence of the angle between the shielding gas flow and the scanning direction on the removal efficiency [131]. (e1) Morphology and size of solidified spatter [132]. (e2) Relationship between spatter number and surface roughness [133]. (e3) Numerous porosity defects induced by spatter [134]. (e4) Quality degradation caused by oxygen-rich spatter falling back onto the scan track [135].

between 0° and 30° (average approximately 15.9°) [138]. Thus, spatter dynamics are fundamentally determined by material thermophysical properties, vapour impingement direction, and jet intensity.

Once airborne, spatter trajectories are modulated by the shielding gas flow and the laser scanning strategy. The shielding gas provides a background flow that entrains and transports most particles. Trajectories and landing positions depend strongly on particle size and inertia. Large particles (≥80 μm) with long relaxation times couple weakly to the gas and exhibit inertia-dominated motion, landing closer to the scan track (Figure 9(d1)) [74].

High-speed imaging with three-dimensional particle tracking reveals that spatter paths are not simple parabolas but exhibit compound oscillation and drift, modulated by the periodic oscillation of the vapour plume. The final deposition and its impact on the build depend on the orientation of the scanning path relative to the gas flow. When scanning is opposite to the gas flow (upstream scanning), spatter is carried backward over the already scanned region, reducing contamination of the fresh powder ahead [74, 131, 134]. Simulations and experiments confirm that upstream scanning induces a two-stage reversal-and-return motion, leading to dense spatter deposition on and downstream of the component [74]. Conversely, downstream scanning blows spatter onto the un-scanned area, creating detrimental residual spatter. Studies further quantify the removal efficiency, finding that it increases with the angle between the scanning direction and the gas flow, peaking at 180° (fully upstream) (Figure 9(d2)) [131]. This directional sensitivity highlights the potential to control spatter trajectories through process planning. The initial ejection angle (mean approximately 15.9° upward) is significantly deflected by the transverse shielding gas, so the final spatial distribution reflects the combined effect of initial dynamics and the external flow field.

5.3.3. Detrimental effects of spatter on building quality

The detrimental effects of spatter on the build quality of the PBF-LB process have been examined using in situ monitoring and ex situ validation methods. The oxide layer on the surface of spatter particles can reach a thickness of several tens of nanometres, far exceeding that on the original powder. When such highly oxidised spatter is redeposited onto the powder bed, it locally increases the powder layer thickness (see Figure 9(e1)) and impedes heat conduction, causing insufficient energy to fully melt the region during subsequent laser scanning and thereby forming lack-of-fusion defects at the melt pool boundary [132, 133, 139]. As shown in Figure 9(e4), the oxygen content on the spatter surface can be 5–8 times that of the original powder, and the oxide layer is enriched with oxides of reactive elements such as aluminum and magnesium. These oxides are difficult to remove in the subsequent melt pool and act as primary initiation sites for lack-of-fusion defects [140].

Comparison of in situ optical tomography with X-ray computed tomography showed that approximately 79% of the lack-of-fusion defects are associated with spatter redeposition, with an average distance of about 0.35 mm between the spatter detection point and the defect centre, showing a strong association between spatter and lack-of-fusion defects [139]. Further spatial statistical analysis showed that the locations of lack-of-fusion defects depend significantly on the positions of spatter particles, whereas the positions of the spatter particles are not influenced by the defect locations, supporting spatter as a cause rather than a consequence of defect formation [141]. The association is stronger for larger defects. The larger the defect, the more spatter signals are detected in its vicinity, and the matching probability between spatter and defect increases monotonically with size. When the spatter diameter exceeds 770 μm, the matching rate reaches 100%, indicating that large spatter particles are the primary cause of random large-sized lack-of-fusion defects.

The correlation mechanism between spatter and gas pores has also been examined. Synchrotron X-ray imaging studies revealed that during the fallback of spatter, micron-scale gaps often exist between the trapped spatter particles and the substrate. These gaps are compressed by thermal expansion during subsequent heating of the scan track, but the gas cannot be completely expelled and can subsequently form spherical gas pores [17]. This mechanism explains why some gas pores do not originate from keyhole collapse but instead coincide closely with spatter fallback positions. Figure 9(e3) shows the characterisation of pore defects in two experimental blocks with a large difference in spatter events, revealing that the porosity under severe spatter conditions is far greater than that under low-spatter conditions [134].

Spatter also affects later layers by degrading the surface of the current layer. High-speed imaging combined with fringe projection profilometry shows a positive relationship between the amount of spatter and the roughness of the deposited layer (Figure 9(e2)) [133]. A rough surface can weaken interlayer bonding and increase the likelihood of lack of fusion, gas pores, and cracks in subsequent layers. However, not every spatter particle produces a defect. Small particles may

be completely remelted or removed by the recoater, and larger particles that are weakly bonded to the surface may also be removed during recoating [141].

5.4. Denudation phenomenon

Denudation appears as belt-like zones along both sides of the laser scan track where powder has been cleared. Unlike spatter, denudation does not directly generate new molten material but instead alters the local packing state of the powder bed and therefore affects scan track geometric integrity and metallurgical bonding quality. Spatter concerns mass loss from the melt pool, whereas denudation involves changes in the powder bed environment. Although both are driven by the vapour jet, their respective pathways of impact on build quality differ.

5.4.1. Phenomenon and driving forces

In PBF-LB, the denudation zone is a powder-cleared belt along both sides of the laser scan track, formed by multiple physical forces (Figure 10(a)). Under atmospheric pressure, the metal vapour jet induces a shear flow in the ambient gas through the Bernoulli effect, and the inward entrainment of powder particles toward the melt pool is the dominant denudation mechanism. A fraction of particles is captured by the melt pool and the remainder is ejected away [62]. As ambient pressure decreases, the gas entrainment effect first intensifies and then weakens. At approximately 4 Torr, it balances the direct momentum transfer from the vapour, producing the minimum denudation zone width. Below this pressure, direct particle ejection by the vapour dominates [143]. Multiphase flow simulations confirm that the metal vapour jet governs both denudation and spatter, while thermal buoyancy is negligible. The vapour jet velocity decays sharply along the ejection direction and expands radially, inducing a pronounced vortex in the ambient gas that entrains particles inward (Figure 10(b1,b2)) [137]. A coupled CFD-DEM model reveals that the vapour ejection angle significantly modulates the denudation zone width, with backward ejection producing the strongest entrainment and the widest zone, followed by forward ejection, and vertically upward ejection giving the weakest effect. The vortex structure induced by the vapour jet constitutes the core driving force for powder entrainment [143].

In addition to external environmental forces, the denudation phenomenon is also governed by various intrinsic forces. Analyzed from the perspective of particle-scale force balance, denudation results from the competition among drag force, friction force, van der Waals adhesion force, and gravity. In fine powders, adhesion forces dominate, leading to a clearly defined denudation zone boundary accompanied by agglomerates. In coarse powders, gravity dominates, resulting in a blurred boundary and dispersed particles. The denudation width exhibits a non-monotonic variation with particle size, reaching a maximum where the adhesion force and gravity are comparable [146].

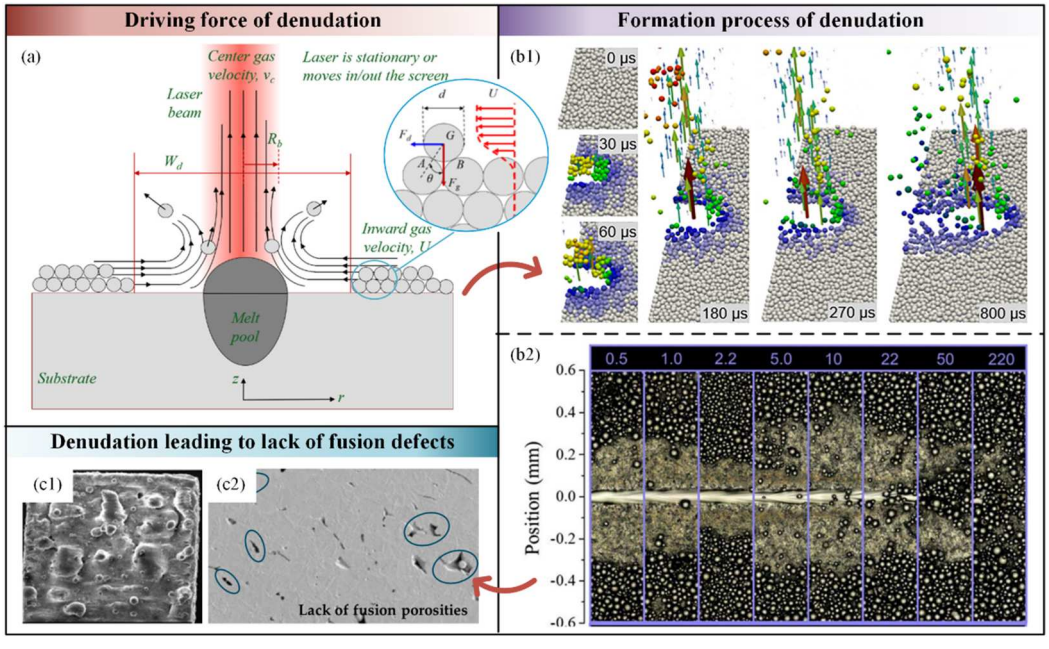


Figure 10. (a) Analysis of driving forces during denudation formation [142]. (b1) Denudation zone formation induced by vapour jetting [143]. (b2) Image of denudation zone in experiments [62]. (c1) Surface morphology of a block affected by denudation [144]. (c2) Lack-of-fusion defects caused by denudation [145].

These studies collectively indicate that the denudation phenomenon is essentially the outcome of the combined action of the ambient gas dynamics induced by the metal vapour jet and the particle-scale mechanics. Its intensity is modulated by multiple factors, including the ambient pressure, vapour ejection angle, powder particle size, and material properties.

5.4.2. Influence of denudation on build quality

Denudation significantly affects build quality through scan track geometry, surface roughness, pore formation, and interlayer bonding. Variations in denudation zone width directly influence scan track formation. Under atmospheric pressure, gas entrainment effectively transports powder into the melt pool, yielding a scan track height of approximately 30 μm, whereas low pressure pushes powder outward and reduces the height to approximately 11 μm, indicating that denudation alters powder replenishment efficiency and scan track geometric integrity [62]. High-speed camera observations further show that partially melted powder entrained in the melt pool impacts the surface in a semi-solid state, becoming an important source of surface roughness.

Denudation is deeply coupled with melt pool hydrodynamics and influences defect formation. High-fidelity simulations indicate that entrainment of incompletely melted powder particles preserves inter-particle pores, leaving shallow pores of approximately 5 μm at the side of the scan track [52]. Synchrotron X-ray imaging has directly observed this process and found that subsequent scan tracks can partially remelt these pores, but insufficient remelting transforms them into stable lack-of-fusion defects [147] (Figure 10(c1, c2) [144, 145]).

Hatch spacing governs the cumulative effect of denudation and its ultimate impact on build quality. Coupled CFD-DEM simulations show that at hatch spacings of 140 μm and 210 μm, the subsequent scan track enters a powder-depleted region where the powder volume fraction drops to 21.1% and 21.9%, causing discontinuous melting and lack-of-fusion defects. When the hatch spacing increases to 350 μm, the denudation effect weakens, the powder volume fraction recovers to 35.4%, and scan track continuity improves, although more spatter particles fall back [148]. The powder replenishment coefficient, defined as the ratio of actual to nominal powder bed volume fraction, directly correlates with scan track discontinuity, with lower values increasing the susceptibility to discontinuous defects [52].

Laser energy distribution indirectly modifies how denudation influences build quality by altering metal vapour flow. Under a double-ellipsoid distribution, the radial ejection and inward entrainment velocities (3.3 m/s and −2 m/s) are significantly lower than those for Gaussian and uniform distributions, producing the narrowest denudation zone (∼782 μm) and the lowest spatter fraction (∼1.6%) [71]. In contrast, a Gaussian distribution exhibits a large vapour temperature gradient and unstable ambient gas flow, leading to the widest denudation zone (∼932 μm) and more pronounced powder bed damage from falling spatter particles, although the spatter ratio is approximately 1.5%. Experimental validation confirms that under the double-ellipsoid distribution, the denudation zone width deviates by only 3.7% from simulations and the spatter angle of ∼56° closely matches predictions, indicating that this energy distribution reduced denudation and spatter under the tested conditions.

6. Defect mitigation methods

Improved understanding of defect formation has led to strategies that act either at the source of the instability or during processing. These strategies include process-window selection, beam shaping, temporal power modulation, preheating, environmental control, and external-field assistance. None is beneficial under all conditions. A change that stabilises one phenomenon may also cause insufficient fusion, excessive evaporation, texture changes, powder disturbance, additional hardware requirements, or a narrower operating window. The following subsections discuss both the reported benefits and the practical limitations of each approach.

6.1. Adjustment of process parameters

Defining a safe process window requires accurate identification of key energy thresholds, typically starting with empirical formulas based on volumetric energy density. This single scalar parameter integrates laser power, scan speed, layer thickness, and hatch spacing and is widely used for preliminary screening because of its simplicity. However, as a design parameter, volumetric energy density has significant limitations. At the same volumetric energy density, different combinations of power and speed yield distinct scan track morphologies and melt pool modes, indicating that this parameter cannot represent underlying mechanisms such as Marangoni convection and recoil pressure [149]. For specific material systems, volumetric energy density can still suggest a process window. In pure tungsten, the relative density initially increases and then decreases with volumetric energy density, and a near-fully dense part is obtained at 1000 J/mm^3 [150]. This points to an optimal energy density window below which lack-of-

fusion pores dominate and above which gas pores increase from melt pool overheating.

Single-track experiments provide a more precise means of identifying boundaries by observing the evolution of scan track morphology and combining melt pool morphology with flow patterns. Cunningham et al. [3] used ultra-high-speed synchrotron X-ray imaging to reveal the real-time transition from conduction mode to keyhole mode and pinpointed the keyhole formation threshold at approximately 0.4 MW/cm^2. They further proposed the keyhole front wall angle as a stability criterion, noting that when the angle exceeds 77° instability and pore generation become likely (Figure 11(a)).

Building on the morphological boundaries from single-track tests, bulk experiments are needed to establish a process window suitable for three-dimensional components. As shown in Figure 11(b), conventional volumetric energy density does not identify the low-porosity window. A modified model that incorporates material absorptivity and thermal diffusivity confines parameter combinations yielding porosity below 1% to a narrow range [151]. Another approach constructs a defect structure process map, dividing the laser power-scan speed space into lack-of-fusion pore, keyhole pore, and nominally fully dense regions. The lack-of-fusion boundary can be identified by a melt pool geometric overlap criterion (Figure 11(c)), and the keyhole boundary by a front wall angle model, outlining a clear window in the three-dimensional space of power, speed, and hatch spacing [152].

Data-driven approaches can reduce the number of experiments needed to identify a process window. Machine learning models learn nonlinear mappings between process parameters and build quality from existing data, reducing reliance on repeated trial-and-error. For PBF-LB, a Bayesian multi-task model that introduces physical laws as prior knowledge extracted critical geometric conditions for avoiding interlayer lack-of-fusion defects, specifically a melt pool depth-to-layer-thickness ratio above 1.62 and a hatch spacing to melt pool width ratio below 0.76 [153]. A stacked ensemble combining artificial neural networks and gradient boosting regression improved prediction accuracy for Ti6Al4 V

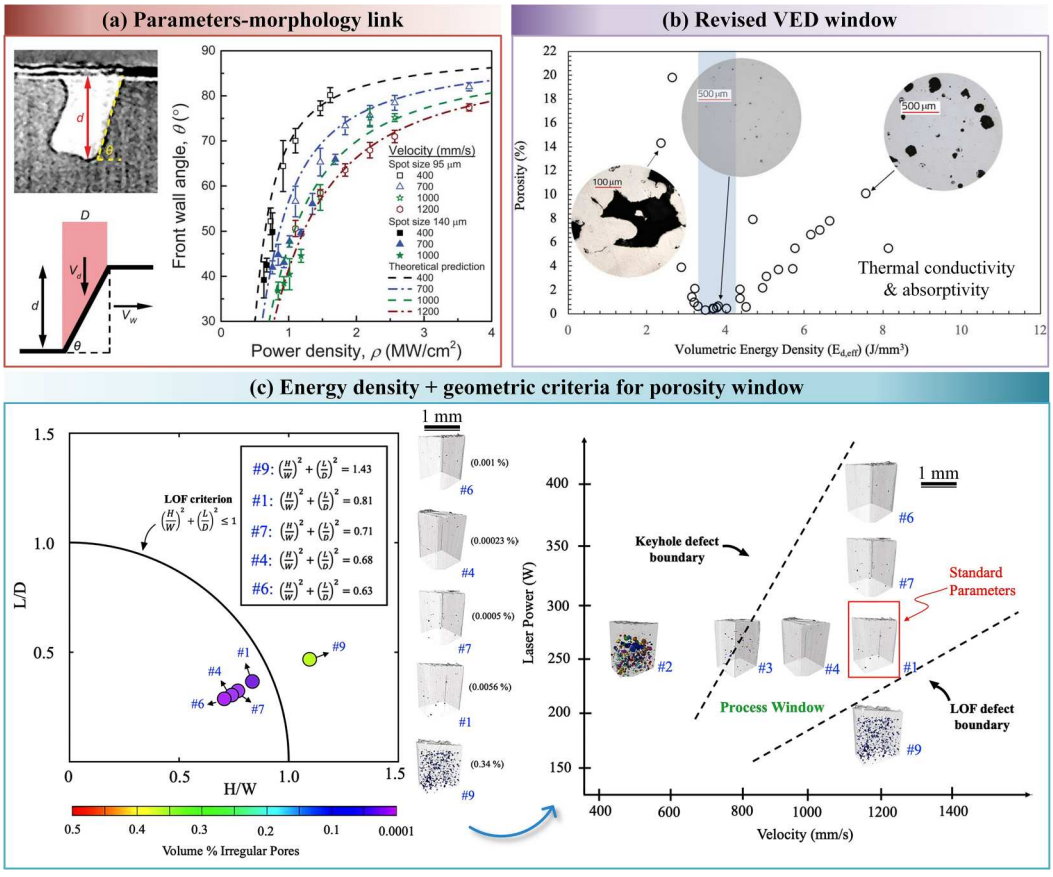


Figure 11. (a) Relationship between keyhole instability and keyhole front wall angle [3]. (b) Process window for low-porosity parts delineated using a modified volumetric energy density model [151]. (c) Identification of the lack-of-fusion boundary using a melt pool geometric overlap criterion and division of the process map in the laser power versus scan speed space into a lack-of-fusion pore region, a keyhole pore region, and a nominally fully dense region [152].

alloy properties [154]. Data-driven methods can explore high-dimensional parameter spaces and reveal complex interactions, although their reliability depends on training data scale and quality.

Online monitoring and closed-loop control allow parameters to be adjusted within a validated operating range during a build. Coaxial melt pool vision or thermal imaging extracts real-time morphological and thermal features and can be used to adjust laser power or scan speed according to preset quality criteria [155]. A digital twin integrating physics-based simulation, data-driven learning, and in situ sensing feedback may estimate internal states and defect risk during a build to support feedback control. Such control may compensate for process drift caused by heat accumulation or changes in the powder bed, but its performance must be verified across full-scale components.

6.2. Beam and energy control

Spatial and temporal beam shaping changes how laser energy is delivered to the powder bed. It can suppress keyhole instability and spatter by controlling peak intensity, melt pool geometry, and heat accumulation. The outcome nevertheless depends on the selected beam profile, modulation parameters, material, and geometric feature.

6.2.1. Beam shaping technology

Conventional Gaussian beams concentrate energy at the spot centre, easily causing localised overheating, keyhole instability, and spatter. Spatial beam shaping modifies the transverse energy distribution, offering new degrees of freedom for controlling melt pool dynamics and suppressing defects. Existing techniques can be categorised into three levels, namely static shaping, dynamically programmable control, and numerically optimised design.

Static shaping includes schemes such as flat-top, ring, and Bessel beams (Figure 12 (a1, a2)). The flat-top beam, with its uniform energy distribution, significantly reduces peak temperature and increases cooling rate. In 316L stainless steel fabrication, it produces a stable melt pool with a width-to-depth ratio below 0.5 and a build rate 14.3 times that of a conventional Gaussian beam [162]. Ring beams distribute energy to the spot periphery and suppress gas pores and spatter by controlling keyhole morphology. A relative density of approximately 99.97% is achievable in Ti-6Al-4 V alloy, yet the process window is limited and the central protrusion at high power can break up into spatter [160]. Bessel beams possess non-diffracting and self-healing properties, and their concentric ring-shaped energy distribution effectively reduces the peak temperature and temperature gradient at the melt pool centre. They significantly suppress keyhole-induced pore defects and spatter while imparting greater defocus tolerance and process robustness. Coupled thermal-microstructure simulations reveal that the Bessel beam produces a smaller melt pool, finer grains after solidification, and a markedly higher proportion of equiaxed grains compared with those fabricated using Gaussian and ring beams.

Dynamically programmable control builds on static shaping by enabling real-time adaptive switching of the beam shape, adding process flexibility for zonal printing of complex components. In situ monitoring captures signals such as melt pool thermal radiation, morphology, and spatter in real time, and adjusts the beam energy distribution according to preset process objectives, thereby stabilising the melt pool state [163, 164]. As shown in Figure 12(b), stepless dynamic switching among Gaussian, ring, and point-ring hybrid beam modes allows adaptive energy allocation for different process stages and component features [160]. The all-electric beam shape adjustment can be completed within a sub-second and requires no manual replacement of optical components.

Recent beam-shaping studies have moved beyond preset static profiles to numerically optimised intensity distributions. An adjoint-based optimisation framework, presented in Figure 12(c), inversely computes the required beam intensity distribution from a desired melt pool temperature distribution. In an optimisation targeting a uniform melt pool, the resulting dual-peak ring beam increased melt pool width by 12%, reduced depth by 37%, and lowered the maximum temperature from nearly 6000 °C under a Gaussian beam to about 3000 °C, substantially reducing overheating [161]. Dynamic beam shaping based on liquid crystal spatial light modulators is advancing rapidly, and when combined with AI-driven modulation pattern calculations, it could enable more complex and precise beam control.

6.2.2. Temporal energy modulation techniques

Temporal energy modulation varies laser power during scanning to modify periodic melt pool and keyhole dynamics. The effects on build quality, microstructure, and defect formation depend on the modulation timescale.

Laser-material interaction in PBF-LB is inherently dynamic and non-steady-state. Temperature, flow, and keyhole morphology evolve over microseconds to milliseconds. Under continuous wave (CW) mode, constant power input often leads to keyhole instability and pores because of the competition between vapour

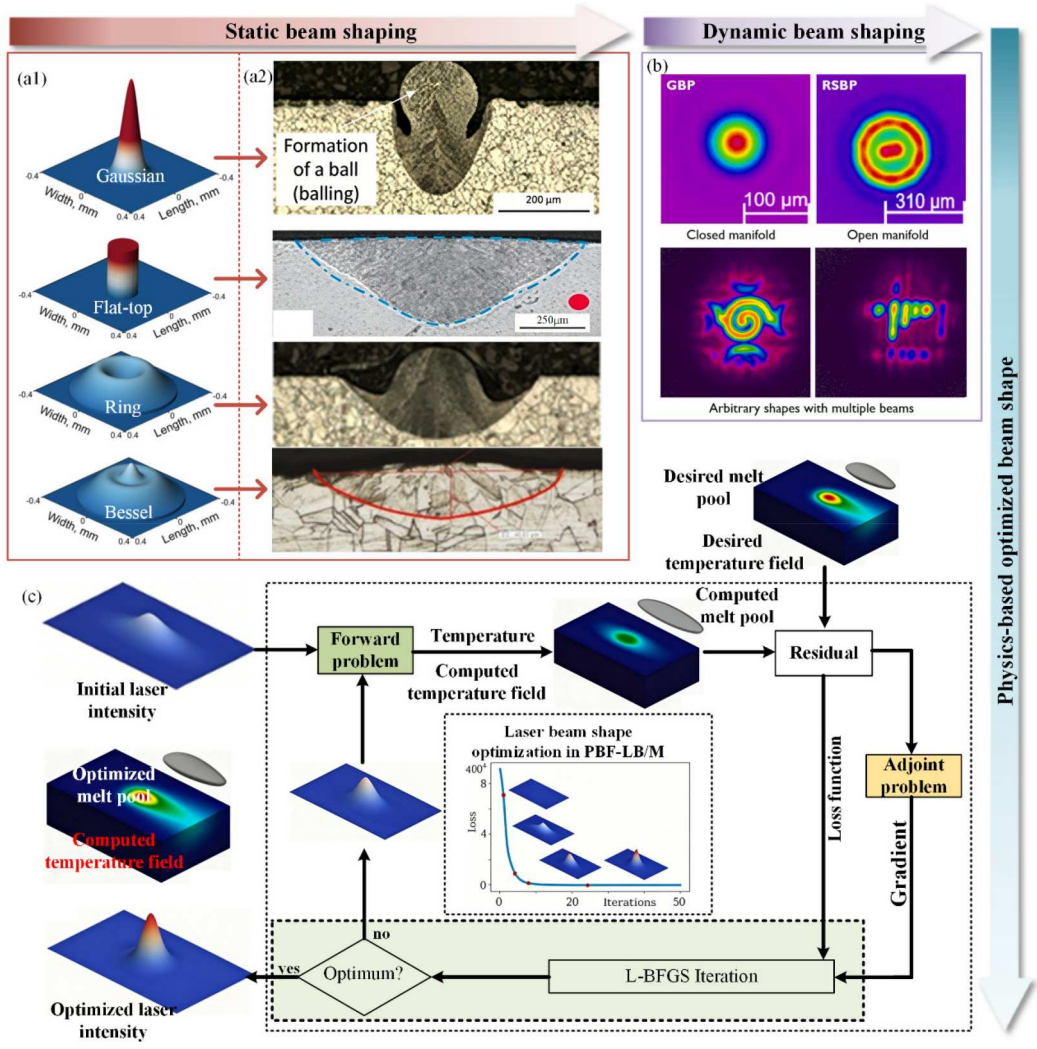


Figure 12. (a1) Four different beam profiles, including Gaussian, flat-top, ring, and Bessel [156]. (a2) Melt pool morphologies under four different beams, including Gaussian [157], flat-top [158], ring [157], and Bessel [159]. (b) Dynamic adjustment of the beam to adapt to different working conditions [160]. (c) Spot shape designed through physics-based numerical optimisation [161].

recoil pressure and surface tension. Temporal modulation uses a non-steady-state laser-power waveform with a specified frequency, duty cycle, and shape to modify melt pool evolution.

Early investigations with millisecond- to microsecond-scale pulses first revealed the influence on build precision and microstructural evolution. A study comparing CW and pulsed wave modes in Ti64 thin-wall fabrication [165] found that intermittent energy input in the pulsed mode reduced heat accumulation, produced finer α′ martensitic needle-like structures, and maintained better angular accuracy at high inclination angles. However, the pulsed mode caused dimensional deviation at low angles due to gravitational effects, whereas the CW mode resulted in coarser microstructures and more powder adhesion (Figure 13(a)). This demonstrated the feasibility of regulating heat input and cooling rate through laser on-off switching.

With higher modulation frequencies, richer physical effects emerged. High-frequency modulation at 50 kHz for Inconel 625 reduced grain size by 30-40% and altered crystallographic texture, attributed to a self-induced ultrasonic effect from minute melt pool oscillations. However, it also narrowed the stable process window and slightly increased porosity [167].

A laser keyhole-reshaping method has been proposed to reduce keyhole instability. This method coaxially combines a CW laser with a nanosecond pulsed laser. The pulsed-laser-induced shock wave exerts an instantaneous outward pressure on the keyhole wall, dynamically counteracting surface tension and other collapse-driving forces. The combined beams transformed the keyhole morphology from a collapse-prone J-shape to a stable I-shape (Figure 13(b)), substantially reducing aluminum alloy porosity from 3.63% to 0.14% while refining grains and enhancing mechanical properties [166].

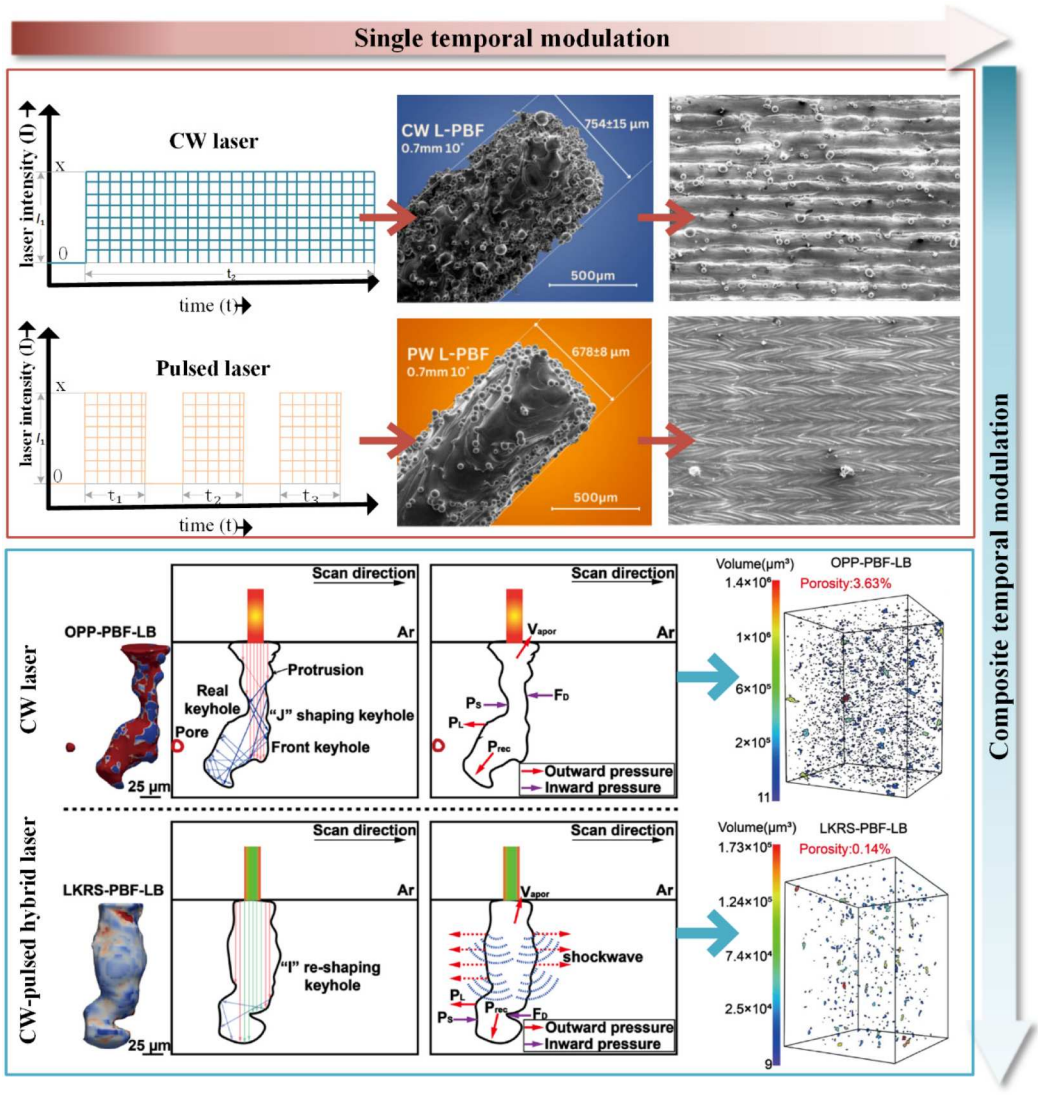


Figure 13. (a) Use of a pulsed laser to reduce particle adhesion on the scan track surface, thereby improving surface quality [165]. (b) Combination of continuous wave and pulsed lasers to regulate keyhole shape, thus reducing part porosity [166].

In situ high-speed X-ray imaging studies have further quantified the relationship between modulation parameters and defect formation [168]. In the intermediate duty cycle regime, keyhole collapse at laser switch-off was identified as the primary cause of gas pore generation, even allowing the controlled creation of one subsurface pore per laser pulse. Under low duty cycles, the melt pool fully solidified between pulses, and the laser repeatedly acted on solid substrate, producing independent melt tracks and vertically elongated pores.

Temporal energy modulation has developed from simple on-off switching to multiscale strategies that include pulse shaping and high-frequency modulation and may ultimately incorporate feedback control. These approaches may improve process stability and provide more precise control of microstructure and defects in PBF-LB.

6.3. External field assistance and environmental control

External physical field assistance (such as ultrasonic or magnetic fields) and the control of environmental parameters together with powder functionalization are both effective approaches for regulating melt pool behaviour and solidification structures. The former acts on melt flow and crystallization through non-contact force fields, while the latter optimises the processing environment and feedstock characteristics. Together they constitute a systematic strategy that extends from external intervention to internal material control and from process conditions to the material itself.

6.3.1. External energy field assistance

External energy fields modify melt pool flow and solidification without changing the nominal alloy composition

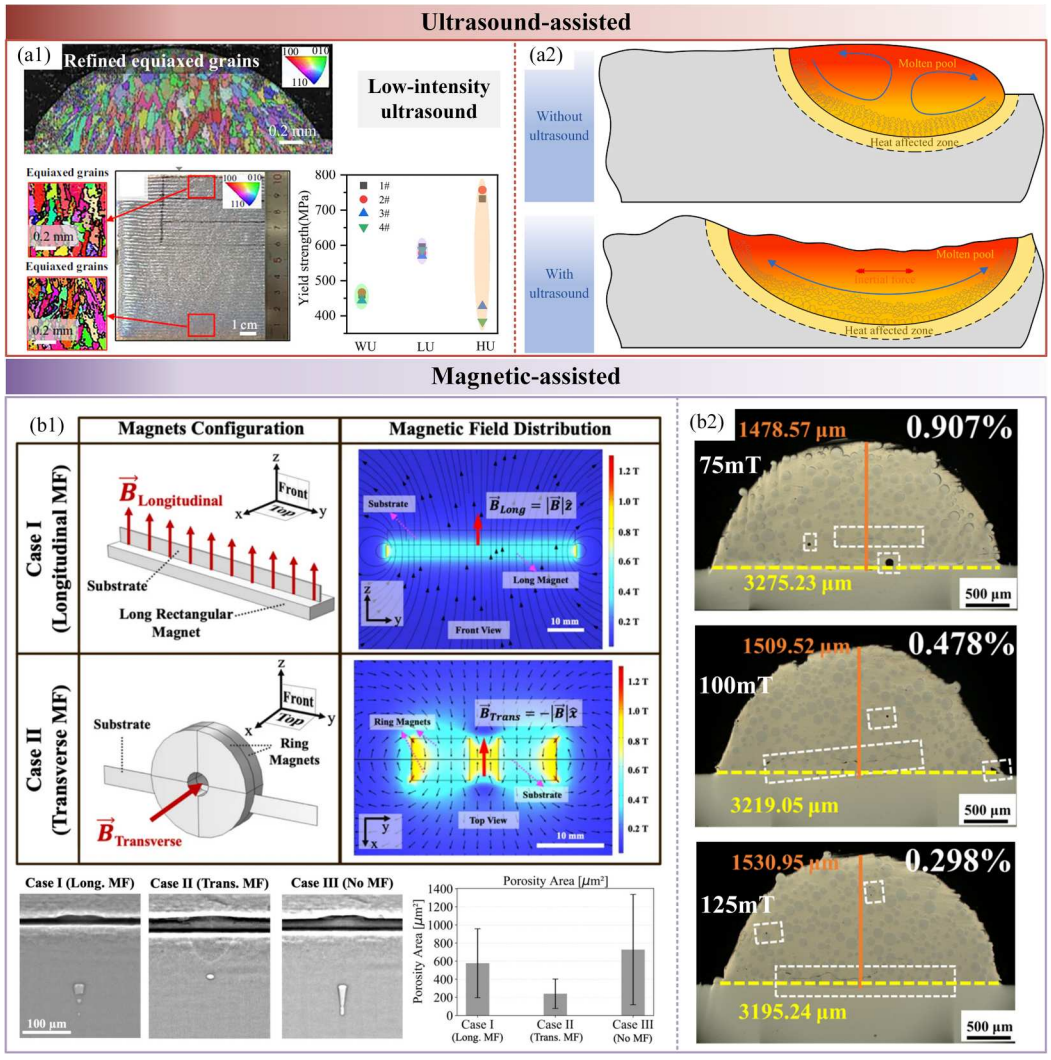


Figure 14. (a1) Grain refinement achieved under pure acoustic streaming [169]. (a2) Inertia-force-dominated reciprocating flow pattern induced in the melt pool by ultrasound [170]. (b1) Influence of transverse magnetic field, longitudinal magnetic field, and no magnetic field on pore defects [175, 176]. (b2) Influence of magnetic field strength on the suppression of defects such as gas pores and cracks.

and can therefore be used to address coarse microstructures, elemental segregation, and process defects.

Ultrasonic assistance couples vibrations above 20 kHz into the melt pool through acoustic streaming and cavitation. Conventional theory considers cavitation-induced shock waves as the main source of dendrite fragmentation and nucleation, while acoustic streaming was long regarded as secondary due to its low velocity [169, 170]. This view has been revised by a noncontact low-intensity ultrasound technique in which the transducer moves synchronously with the laser nozzle and transmits waves below 17.5 W/cm^2 through gas, exposing the melt pool solely to acoustic streaming. As shown in Figure 14(a1), acoustic streaming alone produced measurable grain refinement [169]. Another study confirmed that an inertia-force-dominated reciprocating flow refined grains, promoted columnar-to-equiaxed transition, and reduced the average pore diameter from 45 μm to 18 μm (Figure 14(a2)) [170].

El-Azab et al. found that ultrasonic vibration reduced porosity by 92% and dendrite arm spacing by 54% near the substrate during directed energy deposition of 316L [171]. High-speed X-ray imaging showed that ultrasonic treatment can alter the Al6061 keyhole shape, reduce pores caused by keyhole collapse, and decrease dynamic instability [172]. The outcome depends strongly on the ultrasonic parameters, particularly the intensity [173]. Inertial and Marangoni forces compete to determine melt circulation and bubble escape. Optimised frequency and amplitude can eliminate gas pores, whereas settings outside this range may leave the porosity unchanged or introduce new defects [174]. Very high

acoustic intensity in a contact configuration, for example 849.6 W cm-2, can refine the grains but also greatly increase porosity and surface damage [169].

Magnetic-field assistance uses the Lorentz force generated by the interaction between an applied magnetic field and the electrically conductive melt. Under thermoelectric magnetohydrodynamics, the steep laser-induced temperature gradient generates thermoelectric currents through the Seebeck effect. Their interaction with the applied field produces a Lorentz force whose damping and driving components depend on the relative directions of the current, field, and melt flow. The resulting force can suppress selected velocity components or generate secondary convection, thereby modifying the temperature distribution, keyhole dynamics, solute transport, and solidification behaviour [175–177].

The direction and strength of the magnetic field determine its effect on the melt pool and build quality [175]. Using high-speed synchrotron X-ray imaging, Fan et al. examined how a magnetic field changes keyhole instability in PBF-LB [178]. A protrusion produced by the main vortex at the rear of the melt pool promotes keyhole oscillation, collapse, and pore formation. A transverse magnetic field generates a secondary thermoelectric magnetohydrodynamic flow that changes the vortex structure, suppresses the protrusion, and reduces large-amplitude keyhole oscillations. The effect depends on the orientation of the scan relative to the magnetic field because this orientation determines the Lorentz force generated by the Seebeck current. At the micrometer scale of PBF-LB, electromagnetic damping is relatively weak. For alloys with a large Seebeck coefficient, thermoelectric magnetohydrodynamics becomes the main mechanism by which the field modifies flow behind the keyhole.

In a study on the keyhole-mode laser melting of 316 stainless steel, a longitudinal magnetic field of approximately 120 mT suppressed horizontal flow and deepened the keyhole but intensified lateral fluctuations. A transverse field of approximately 760 mT damped flow in all directions, produced a shallower and more stable keyhole, and reduced pore formation, as shown in Figure 14(b1). Studies of nickel-based tungsten carbide composites have also identified a bounded range of effective magnetic-field strengths. When the magnetic field strength is between 100 and 150 mT, the Lorentz force and the thermoelectric magnetohydrodynamic effect reach a balance, which can suppress gas pores and cracks (see Figure 14(b2)), while promoting the uniform distribution of tungsten carbide reinforcing particles within the melt pool and preventing their segregation [176]. This uniform microstructure promotes the partial dissolution of tungsten carbide particles in the nickel matrix and the precipitation of secondary carbides. The resulting 'core-shell' structure increased the wear resistance of the composite coating by more than 50%.

Ultrasound and magnetic fields modify the melt pool through different mechanisms and within different operating ranges. Their value therefore depends on whether the applied field produces the required flow or solidification response without introducing additional instability, porosity, or equipment constraints.

6.3.2. Environmental control and powder modification

Environmental control changes the thermal and chemical conditions during processing, whereas powder modification changes the composition or surface state of the feedstock. In superalloy additive manufacturing, temperature and loading mode affect microstructure evolution and defect initiation. Studies show that under cyclic loading at 550 °C, fatigue cracks in IN718 propagate transgranularly with no significant microstructural interaction. When the temperature is raised to 650 °C or a dwell time is introduced, the crack growth rate increases sharply, and the path shifts to a mixed transgranular and intergranular mode and even follows secondary dendrite arm boundaries. This behaviour results from the combined effects of temperature and exposure time, which intensify oxidation-induced embrittlement of grain boundaries and substructures, indicating that substrate preheating or chamber temperature control can substantially alter crack propagation and suppress hot cracking [179]. Atmosphere and pressure control are equally critical. In air, stress-accelerated grain boundary oxidation and dynamic embrittlement at the crack tip cause premature failure, highlighting the necessity of vacuum or inert gas protection to suppress environmental embrittlement. Meanwhile, increasing ambient pressure compresses the keyhole, stabilises the melt pool, and reduces shielding gas entrapment and alloying element evaporation, yielding a more uniform and dense deposited layer.

Powder functionalization can be used for microstructural refinement and property enhancement. In Al-Cu-Li-Mg alloys, a trace addition of Mg significantly promotes Cu atom clustering and subsequent precipitate evolution. Mg accelerates Cu diffusion by altering vacancy behaviour, and its effect saturates above 0.2 at.%. When the Mg content exceeds 0.1 at.%, precipitates approximately 2 nm in size form at room temperature and act as heterogeneous nucleation sites for the T1 phase. This mechanism closely parallels the concept of introducing nanoparticles via powder surface coating to increase nucleation sites [179]. Surface modification, such as

Table 5. Benefits, potential adverse effects, and engineering constraints of active defect-mitigation strategies.

Strategy	Intended benefit	Potential adverse effects / failure modes	Critical variables and engineering limitations
Spatial beam shaping	Reduce centreline overheating; control melt pool width and depth; stabilise the keyhole; expand the process window and increase productivity	Insufficient penetration or a cold centre; central protrusions or spatter generation for some ring modes; profile-dependent crystallographic texture	Optical efficiency and calibration; focus/contamination sensitivity; feature-specific profile switching [156–162]
Temporal power modulation	Suppress unstable oscillations; control heat accumulation and keyhole shape	Keyhole collapse during laser switch-off; pore formation; discontinuous tracks; periodic defects; a narrowed process window	Frequency, duty cycle, waveform, synchronisation with scan dynamics, actuator bandwidth [165–168]
Build-platform preheating and chamber-temperature control	Reduce thermal gradient, residual stress, and cracking; improve interlayer bonding	Powder sintering/caking, oxidation or volatile-element loss, grain coarsening, lower cooling rate, dimensional drift	Maximum powder-bed temperature, residence time, atmosphere purity, recoating behaviour, machine thermal management
Ultrasonic assistance	Promote acoustic streaming/cavitation, bubble escape, and grain refinement	Excessive excitation can increase porosity and surface defects; unstable coupling; possible remobilisation of inclusions	Frequency/amplitude window, coupling path, component height/geometry, transducer durability [169–174]
Magnetic-field assistance	Modify melt pool flow through Lorentz forces and thermoelectric magnetohydrodynamics (TEMHD); suppress selected keyhole oscillations and segregation	Direction-dependent keyhole deepening or lateral fluctuation; unintended flow or changes in crystallographic texture; coil heating and electromagnetic interference	Field strength/orientation, scan direction, alloy Seebeck coefficient and conductivity, coil integration [175–178]
Shielding-gas and chamber-pressure control	Remove spatter, limit oxidation, control plume behaviour and denudation, and stabilise vaporisation	Powder-bed disturbance, plume deflection, redeposition or cross-contamination; pressure-control hardware and safety requirements	Velocity uniformity, scan-to-flow angle, gas species, filter loading, part obstruction, sealing and throughput [72–74, 126, 138–141]

coating with oxide nanoparticles, can further adjust laser absorptivity and improve melt pool wettability, enabling active microstructural control. Environmental parameters, including temperature, atmosphere, and dwell time, should therefore be considered together with powder composition and surface condition when process conditions are selected. The microstructural similarity between additively manufactured and welded parts also suggests that powder modification strategies must account for non-equilibrium solidification features such as dendritic segregation and secondary phase distribution to enhance service performance [180]. High-throughput composition-gradient methods allow rapid screening of concentration thresholds for modifying elements and can support quantitative design of powder-modification strategies [179].

6.4. Limitations and trade-offs of active mitigation

Table 5 summarises the adverse effects and implementation constraints discussed in the preceding subsections. Beam shaping can reduce the peak temperature and broaden the melt pool. However, an excessively diffuse or annular profile may reduce penetration or leave a cooler central region, while a high-power ring mode may generate central protrusions and spatter. Temporal modulation can interrupt unstable oscillations, but switching the laser off can itself trigger keyhole collapse. An unsuitable frequency or duty cycle may produce periodic pores, discontinuous tracks, microstructural banding, or a narrower stable window. Preheating reduces thermal gradients and residual stress, but it also increases thermal exposure of the powder bed and machine. Possible consequences include powder sintering or caking, oxidation or evaporation of volatile species, grain coarsening, and dimensional drift.

External fields also have limited operating ranges. Excessive ultrasonic intensity can increase porosity and surface damage. Contact coupling, transducer durability, and attenuation through a growing component further complicate scale-up. Magnetic control depends on the electrical and thermoelectric properties of the alloy and on the orientation of the field relative to the scan and thermoelectric current. A field that damps one flow component may deepen the keyhole or amplify another fluctuation. Stronger gas flow improves spatter removal only until it begins to disturb the powder bed, deflect the plume, accelerate fine particles, or redistribute contamination. Pressure control adds requirements for chamber design, sealing, safety, and throughput. Mitigation studies should therefore report both the optimum conditions and the limits of stable operation, compare cases at matched productivity and energy input, and verify performance in multilayer geometries.

7. Future challenges

Although significant progress has been made in understanding melt pool behaviour and defect-formation mechanisms in PBF-LB, several challenges remain and define key directions for future research. First, with respect to physical mechanisms, existing studies have elucidated the fundamental behaviour of individual phenomena such as Marangoni convection, recoil pressure, and keyhole dynamics. However, quantitative prediction of coupled multiphysics and multiphase interactions remains limited by uncertainties in critical parameters. For example, the temperature coefficient of surface tension, which governs Marangoni convection, is dynamically altered by surface-active elements such as oxygen and sulfur in real alloys. However, in situ measurements of this coefficient remain extremely scarce, resulting in significant deviations in simulated flow patterns and melt pool morphology. Reliable calibration data are also lacking for the evaporation rate coefficient used in recoil-pressure models and for the scattering and absorption coefficients of the vapour plume under different material and energy-density conditions. A unified quantitative description of how the interfacial-instability criterion in keyhole dynamics depends on local curvature, melt viscosity, and pressure gradient has yet to be established. Future work should directly calibrate these parameters through in situ experiments with ultrahigh spatial and temporal resolution, such as synchrotron X-ray imaging and high-speed spectroscopic diagnostics, and establish a multiscale coupling framework spanning nanometres to millimetres and nanoseconds to seconds. Concurrently, the dynamic modification of the surface tension coefficient by surface-active elements and the intrinsic relationship between constitutional undercooling induced by the selective evaporation of low-boiling-point elements and flow instability should be investigated systematically, thereby providing testable quantitative explanations for the physical origins of specific defect types.

Second, in experimental characterisation and data analysis, multiple monitoring techniques can now capture surface morphology, temperature fields, and vapour plumes at high speed, while synchrotron X-ray imaging reveals the internal transient evolution of keyholes and bubbles. Nevertheless, effectively fusing these large, multisource, heterogeneous datasets and extracting features strongly correlated with defect type, size, and spatial distribution remain major bottlenecks in intelligent process monitoring. Existing machine-learning models perform well for specific materials and process parameters but exhibit limited generalizability. When material or equipment conditions change, prediction accuracy often decreases sharply, indicating that current data-driven methods do not sufficiently incorporate the underlying physical mechanisms. Physics-informed machine-learning paradigms should therefore be further developed by using the fundamental laws of thermodynamics, fluid mechanics, and phase-transformation kinetics as soft constraints or as the basis for constructing physically meaningful features. This approach would enable models not only to fit data but also to represent the underlying causal relationships. Meanwhile, advances in multimodal data fusion should enable the synergistic analysis of signals from high-speed cameras, photodiodes, optical microphones, and X-ray systems within a unified framework, providing comprehensive perception of melt pool states and early prediction of defect initiation.

Third, in process control and defect mitigation, methods such as beam shaping, temporal energy modulation, and multiphysical-field assistance have proven effective. However, most studies still focus on validating individual control approaches and lack systematic investigation of the synergistic effects among different mechanisms. More critically, most existing strategies rely on offline-optimised open-loop control, in which a safe process window is predetermined through extensive trial-and-error experiments. Such strategies cannot accommodate dynamic instabilities induced by stochastic disturbances during actual builds, including fluctuations in powder layer thickness, localised heat accumulation, and spatter redeposition. A transition from offline optimisation to online closed-loop control is therefore necessary. This transition requires the close integration of real-time monitoring, intelligent analysis, and multiple-degree-of-freedom actuators to construct an adaptive digital-twin-based control system. Within such a system, machine-learning surrogate models could predict defect risk within milliseconds, while reinforcement-learning controllers could dynamically adjust laser power, scanning speed, beam shape, and even chamber pressure, thereby maintaining the melt pool within the stability range required for consistent formation.

Finally, current research on defect mechanisms is largely confined to single-track, single-layer, or few-layer samples with simple geometries, whereas the formation behaviour of actual parts with complex three-dimensional structures requires further investigation. A considerable body of work has explored the effects of accumulated thermal history, abrupt changes in thermal boundary conditions at overhangs, and interactions among multiple lasers on defect formation. However, several issues remain insufficiently addressed. First, most studies rely on post-process analysis or simplified heat-source models, such as equivalent

volumetric heat sources, making it difficult to capture the two-way coupling between local melt pool dynamics, including keyhole oscillation and bubble migration, and the global thermal history. Second, the quantitative characterisation and control of transient interactions during multi-laser scanning, including vapour plume interactions, spatter cross-deposition, and asynchronous solidification of adjacent melt pools, remain at an early stage. Third, existing defect-sensitivity maps are mostly constructed using constant parameters or simple geometric boundaries and neglect the interactions among structural features, such as thin walls, sharp corners, and overhangs, material response, and spatiotemporal scan-path control. Future research urgently requires computational methods that efficiently couple part-scale thermomechanical evolution with local melt pool dynamics, such as multiscale reduced-order models or machine-learning surrogate models. These methods should be combined with high-throughput experiments and cross-scale in situ characterisation to establish a transferable defect-prediction framework for geometrically complex components.

8. Conclusions

Melt pool behaviour and defect formation mechanisms in PBF-LB involve complex multi-physics, multi-phase, and multi-scale coupling. This paper systematically analyzes the research progress in this field. Advanced in situ characterisation techniques, such as synchrotron X-ray imaging, high-speed optical imaging, schlieren imaging, and acoustic emission, have enabled the observation of transient dynamics of keyholes, melt pool flow, vapour plumes, and spatter at unprecedented spatiotemporal resolution. Multiscale numerical simulations covering macroscopic thermo-mechanical coupling, mesoscopic melt pool dynamics, and microstructural evolution, together with data-driven models such as surrogate models and machine learning, have provided powerful tools for analyzing and predicting the physical mechanisms and quality impacts. On this basis, a causal chain has been established linking melt pool dynamics to vapour plumes, spatter, denudation, and subsequently to defects such as gas pores, lack of fusion, cracks, and segregation. Building on this understanding, various defect mitigation strategies have been developed, ranging from safe process window design, advanced beam spatiotemporal shaping, and multi-physics field assistance to environmental control. Although challenges remain, including incomplete multi-physics coupling mechanisms, limited model generalisation, and the lack of closed-loop control, the deep integration of physical models, data-driven methods, and advanced control theory to construct intelligent digital twin systems is becoming the necessary path toward autonomous quality control and zero-defect manufacturing in PBF-LB.

Author contributions

CRediT: **Haodong Chen:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Jiawei Yan:** Investigation, Visualization, Writing – original draft, Writing – review & editing; **Xinwu Zhang:** Investigation, Writing – review & editing; **Chee Kai Chua:** Investigation, Methodology; **Kunpeng Zhu:** Conceptualization, Investigation, Writing – review & editing; **Xin Lin:** Resources, Supervision, Writing – original draft, Writing – review & editing.

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Data availability statement

No new data were generated or analyzed in this paper.

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